



MOIZ DAWOODI  
O/A LEVEL PHYSICS

# O' LEVEL PHYSICS

## Electrical Circuits

4.3

STUDENT NAME



0315-0805775

PHYSICS WITH  
**MOIZ DAWOODI**

## SYLLABUS OUTLINE:

**4.3 Electric circuits****4.3.1 Circuit diagrams and circuit components**

- 1 Draw and interpret circuit diagrams with cells, batteries, power supplies, generators, oscilloscopes, potential dividers, switches, resistors (fixed and variable), heaters, thermistors (NTC only), light-dependent resistors (LDRs), lamps, motors, ammeters, voltmeters, magnetising coils, transformers, fuses, relays, diodes and light-emitting diodes (LEDs), and know how these components behave in the circuit

**4.3.2 Series and parallel circuits**

- 1 Recall and use in calculations, the fact that:
  - (a) the current at every point in a series circuit is the same
  - (b) the sum of the currents entering a junction in a parallel circuit is equal to the sum of the currents that leave the junction
  - (c) the total p.d. across the components in a series circuit is equal to the sum of the individual p.d.s across each component
  - (d) the p.d. across an arrangement of parallel resistances is the same as the p.d. across one branch in the arrangement of the parallel resistances
- 2 Calculate the combined resistance of two or more resistors in series
- 3 Calculate the combined resistance of two resistors in parallel
- 4 Calculate current, voltage and resistance in parts of a circuit or in the whole circuit

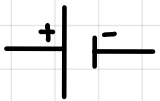
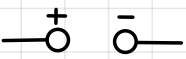
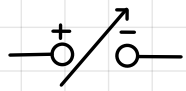
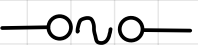
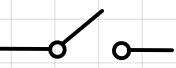
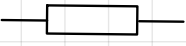
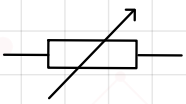
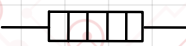
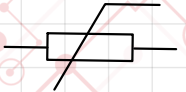
**4.3.3 Action and use of circuit components**

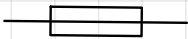
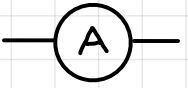
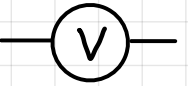
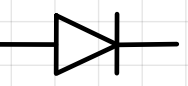
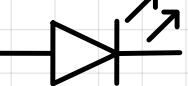
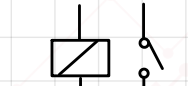
- 1 Describe the action of negative temperature coefficient (NTC) thermistors and light-dependent resistors and explain their use as input sensors
- 2 Describe the action of a variable potential divider
- 3 Recall and use the equation for two resistors used as a potential divider

$$\frac{R_1}{R_2} = \frac{V_1}{V_2}$$



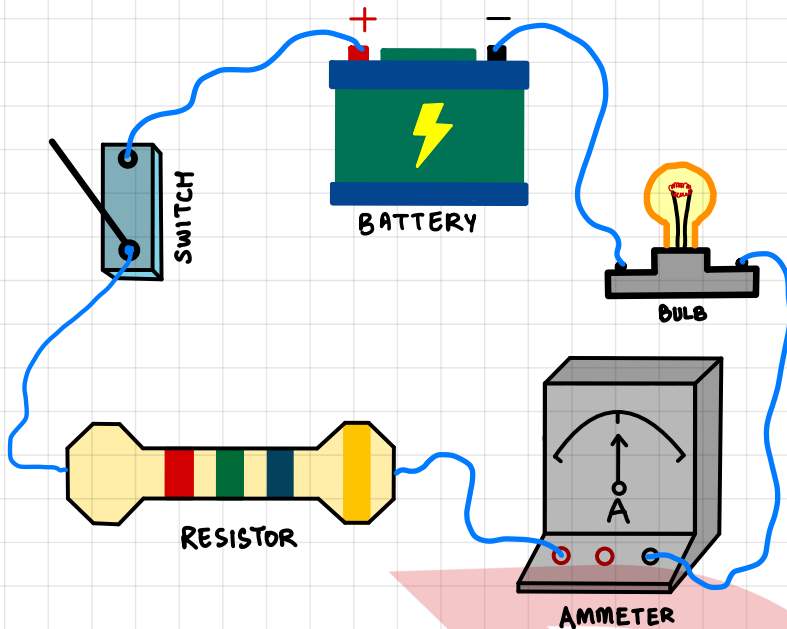
## CIRCUIT SYMBOLS:

| SYMBOL                                                                                          | COMPONENT                      | PURPOSE                                                |
|-------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------------------------------|
|  <sup>DC</sup> | Cell                           | An emf source.                                         |
|  <sup>DC</sup>  | Battery                        | Combination of emf sources                             |
|  <sup>DC</sup>  | Power Supply                   | Converts AC to DC according to the need of appliances. |
|  <sup>DC</sup>  | Variable Power Supply          | Provides variable DC output.                           |
|  <sup>AC</sup> | Mains Supply<br>AC Source      | Provides alternating current.                          |
|              | Open Switch                    | Makes the circuit OFF.                                 |
|              | Closed Switch                  | Makes the circuit ON.                                  |
|               | Fixed Resistor                 | Provides fixed opposition to electric current.         |
|               | Variable Resistor/<br>Rheostat | Provides variable resistance to electric current.      |
|               | Heater                         | Converts electric to thermal energy.                   |
|               | Thermistor                     | A heat sensor.                                         |
|               | Light<br>Dependant<br>Resistor | A light sensor.                                        |

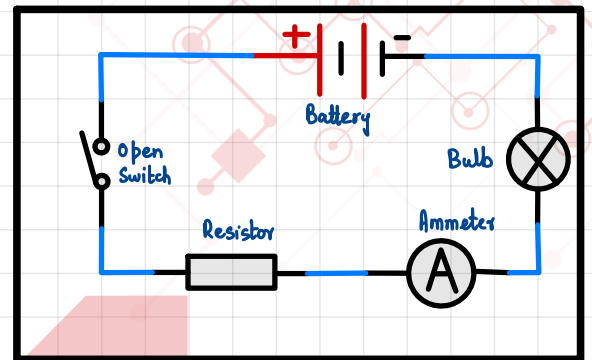
| SYMBOL                                                                              | COMPONENT                   | PURPOSE                                                                 |
|-------------------------------------------------------------------------------------|-----------------------------|-------------------------------------------------------------------------|
|    | Fuse                        | Limit the amount of current                                             |
|    | Ammeter                     | Measures current.                                                       |
|    | Voltmeter                   | Measures voltage & emf.                                                 |
|    | Motor                       | Converts electrical energy to kinetic energy.                           |
|    | Bulb / Lamp                 | Converts electrical energy to light                                     |
|   | Diode                       | Allows current to flow only in one dimension.                           |
|  | Light Emitting Diode        | Diode which glows when it conducts current.                             |
|  | Solenoid / Magnetising Coil | Works as an electromagnet.                                              |
|  | Transformer                 | Converts Alternating Voltages from low to high value or vice versa.     |
|  | Cathode Ray Oscilloscope    | Visualizes waves.                                                       |
|  | Relay                       | A component which switches ON high current appliance using low current. |
|  | Earth Connection            | Connects charged metal to Earth to neutralise it.                       |



# PHYSICAL TO CIRCUIT DIAGRAM:



- Replace components by symbols
- Draw connecting wires at 90°
- Label all components



## 4.3.2

## Series & Parallel Circuits

### SERIES CIRCUIT.

"A circuit in which current has single path to flow."

- Current remains same in series.
- Voltage divides among components in series.

#### SERIES COMBINATION OF RESISTORS:

##### FORMULA:

$$R = R_1 + R_2 + R_3$$

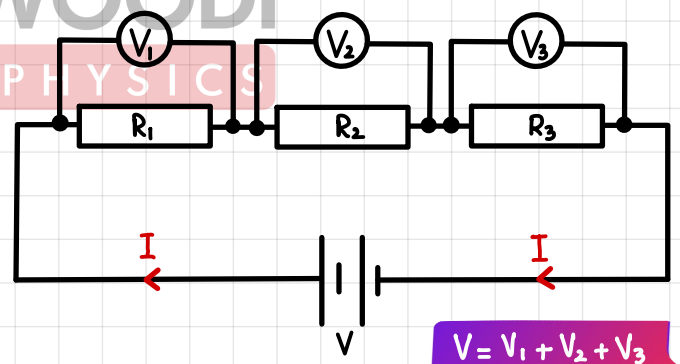
- Series combination of resistors is made when small resistors are given & we want large resistance.

#### EXERCISE:

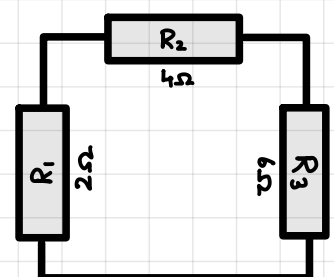
- Calculate the combined resistance of combination.

$$R = R_1 + R_2 + R_3 = 2 + 4 + 6$$

$$R = 12 \Omega$$



$$V = V_1 + V_2 + V_3$$



# PARALLEL CIRCUIT:

"Circuit in which current has more than one path to flow."

- Current divides in parallel.
- Voltage remains same in parallel.

$$I = I_1 + I_2 + I_3$$

## PARALLEL COMBINATION OF RESISTORS:

FORMULA:

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

If two resistors are placed in parallel, then:

$$\frac{1}{R} = \frac{1 \times R_2}{R_1 R_2} + \frac{1 \times R_1}{R_2 R_1} \Rightarrow \frac{1}{R} = \frac{R_2 + R_1}{R_1 R_2}$$

$$R = \frac{R_1 \times R_2}{R_1 + R_2} \quad \text{Product over Sum}$$

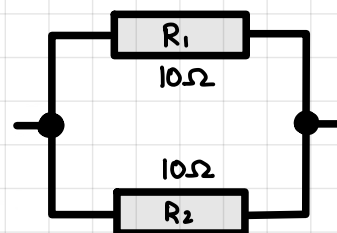
EXERCISE:

- Calculate the combined resistance of combination.

$$R = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{10 \times 10}{10 + 10} = \frac{100}{20}$$

$$R = 5 \Omega$$

If resistance of same value are in parallel then divide resistance of single resistor by no. of resistors. " $R = \frac{R_1}{n}$ "



- Calculate the combined resistance of combination.

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R} = \frac{1}{2} + \frac{1}{4} + \frac{1}{6}$$

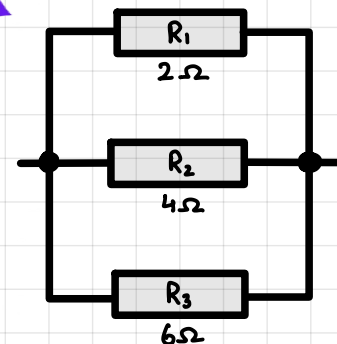
$$\frac{1}{R} = 0.5 + 0.25 + 0.166$$

$$\frac{1}{R} = 0.91$$

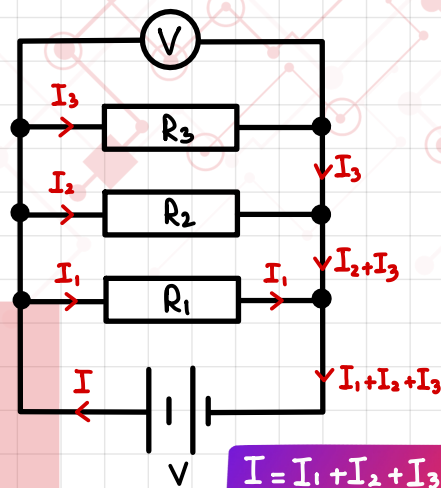
$$R = \frac{1}{0.91}$$

$$R = 1.09 \Omega$$

In parallel, value of equivalent resistance is smaller than smallest resistance. i.e. If  $R_1 = 2\Omega$ ,  $R_2 = 4\Omega$  &  $R_3 = 6\Omega$ . Then  $R < 2\Omega$ .



- We make parallel connection when we have large resistors & we want small resistance

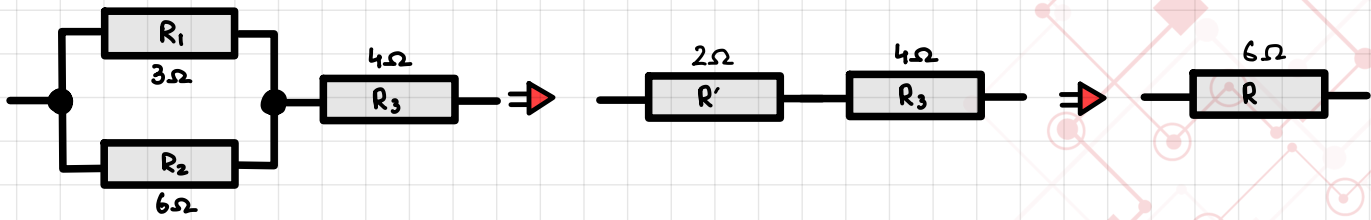


$$I = I_1 + I_2 + I_3$$

# HYBRID COMBINATIONS:

## EXERCISE:

- Calculate the combined resistance of combination.



$R_1$  &  $R_2$  are in parallel:

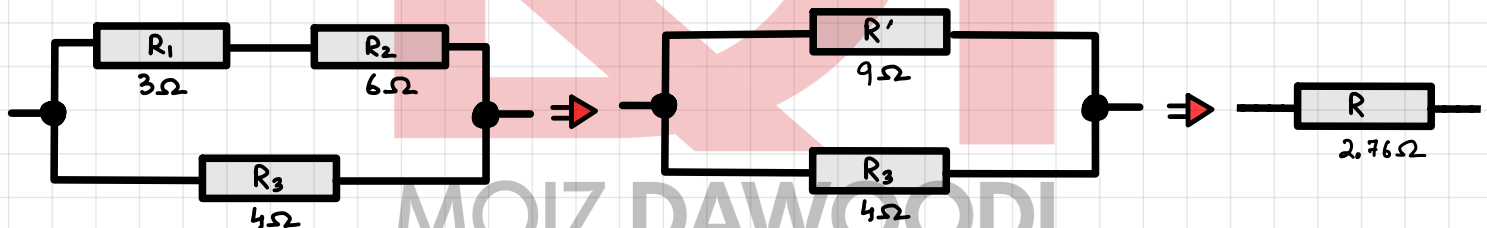
$$R' = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{3 \times 6}{3 + 6} = \frac{18}{9} = 2\Omega$$

$R'$  &  $R_3$  are in series:

$$R = R' + R_3 = 2 + 4$$

$$R = 6\Omega$$

- Calculate the combined resistance of combination.



$R_1$  &  $R_2$  are in series:

$$R' = R_1 + R_2 = 3 + 6 = 9\Omega$$

$R'$  &  $R_3$  are in parallel:

$$R = \frac{R' \times R_3}{R' + R_3} = \frac{9 \times 4}{9 + 4} = \frac{36}{13}$$

$$R = 2.76\Omega$$



# WHOLE CIRCUIT CALCULATIONS:

## EXERCISE:

- From the circuit given, find the following:
- Resistance of two resistors in parallel.

$$R' = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{10 \times 6}{10 + 6}$$

$$R' = 3.75 \Omega$$

- Current  $I$  from the battery.

$$R = R' + R_3 = 3.75 + 12 = 15.75 \Omega$$

$$I = \frac{V}{R} = \frac{5}{15.75}$$

$$I = 0.32 \text{ A}$$

- Potential difference across  $XY$  &  $YZ$ .

$V_{xy}$ :

$$V_{xy} = I R' = 0.32 \times 3.75$$

$$V_{xy} = 1.2 \text{ V}$$

$V_{yz}$ :

$$V_{yz} = I R_3 = 0.32 \times 12$$

$$V_{yz} = 3.8 \text{ V}$$

Alternative Method:

$$5 = V_{xy} + V_{yz}$$

$$5 = 1.2 + V_{yz}$$

$$V_{yz} = 5 - 1.2$$

$$V_{yz} = 3.8 \text{ V}$$

- Current of  $R_1$  &  $R_2$ .

$I_1$ :

$$I_1 = \frac{V_{xy}}{R_1} = \frac{1.2}{10}$$

$$I_1 = 0.12 \text{ A}$$

$I_2$ :

$$I_2 = \frac{V_{xy}}{R_2} = \frac{1.2}{6}$$

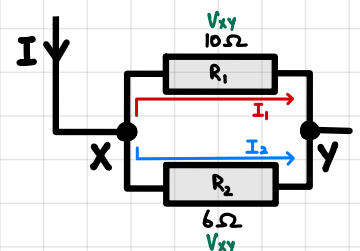
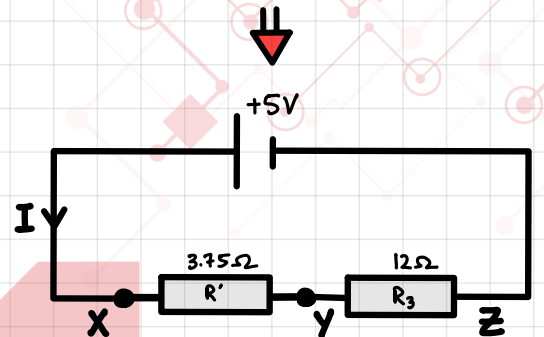
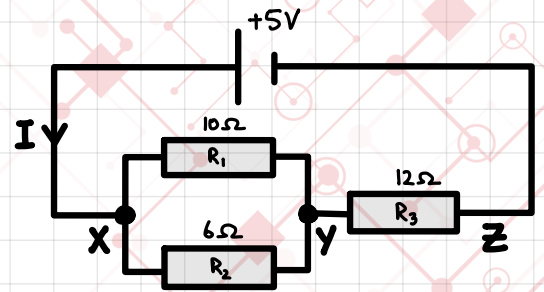
$$I_2 = 0.2 \text{ A}$$

Alternative Method:

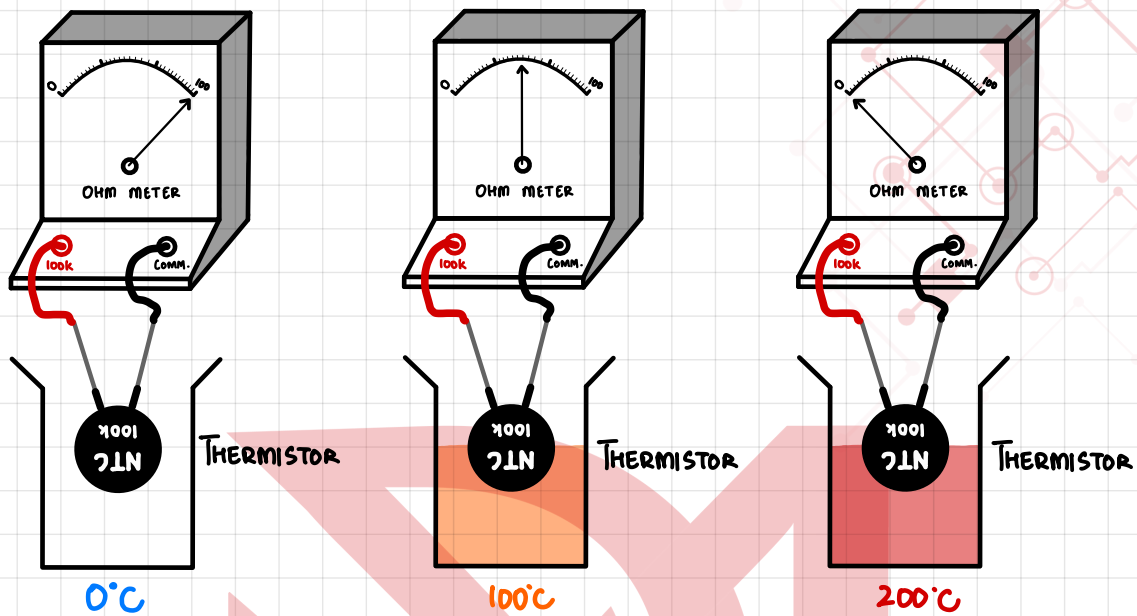
$$I = I_1 + I_2$$

$$I_2 = I - I_1 = 0.32 - 0.12$$

$$I_2 = 0.2 \text{ A}$$



## THERMISTOR:

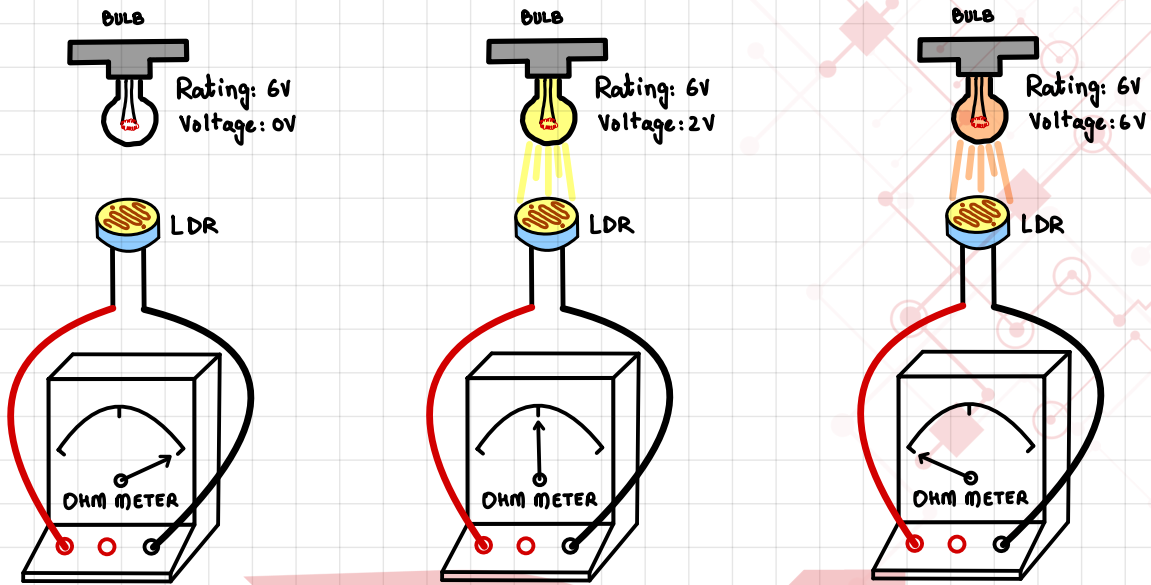


Thermistor belongs to NTC (Negative Temperature Coefficient)  
Semiconductor whose resistance decreases when temperature  
increases or Vice Versa.

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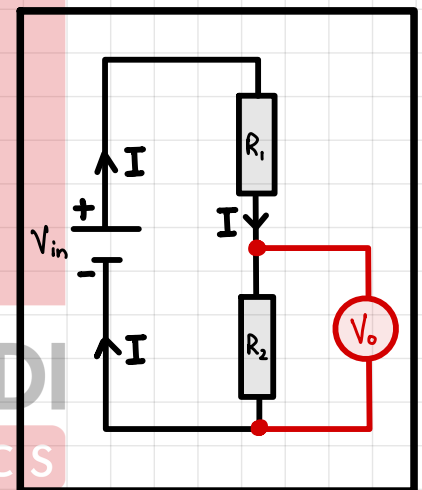


# LDR (LIGHT DEPENDENT RESISTOR):

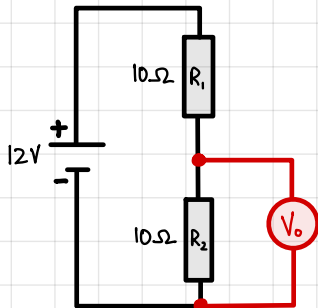
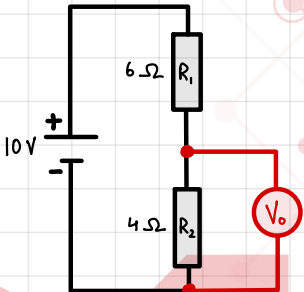
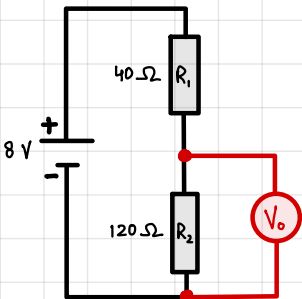
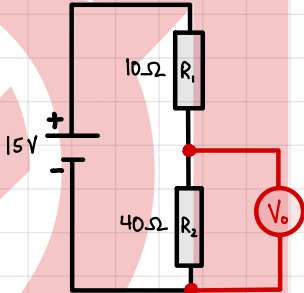


## POTENTIAL DIVIDERS:

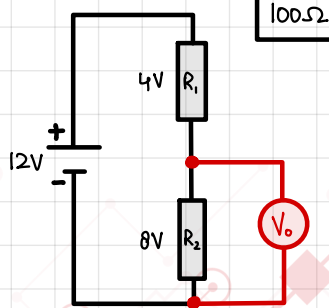
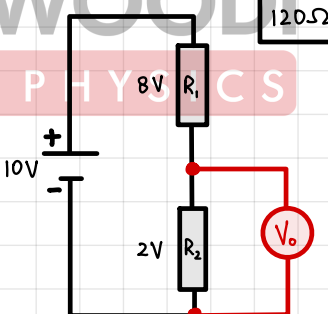
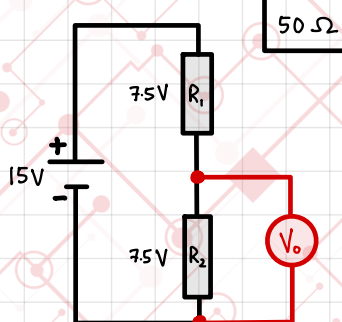
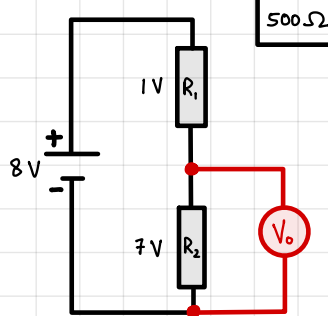
FORMULA:



# RATIO METHOD.

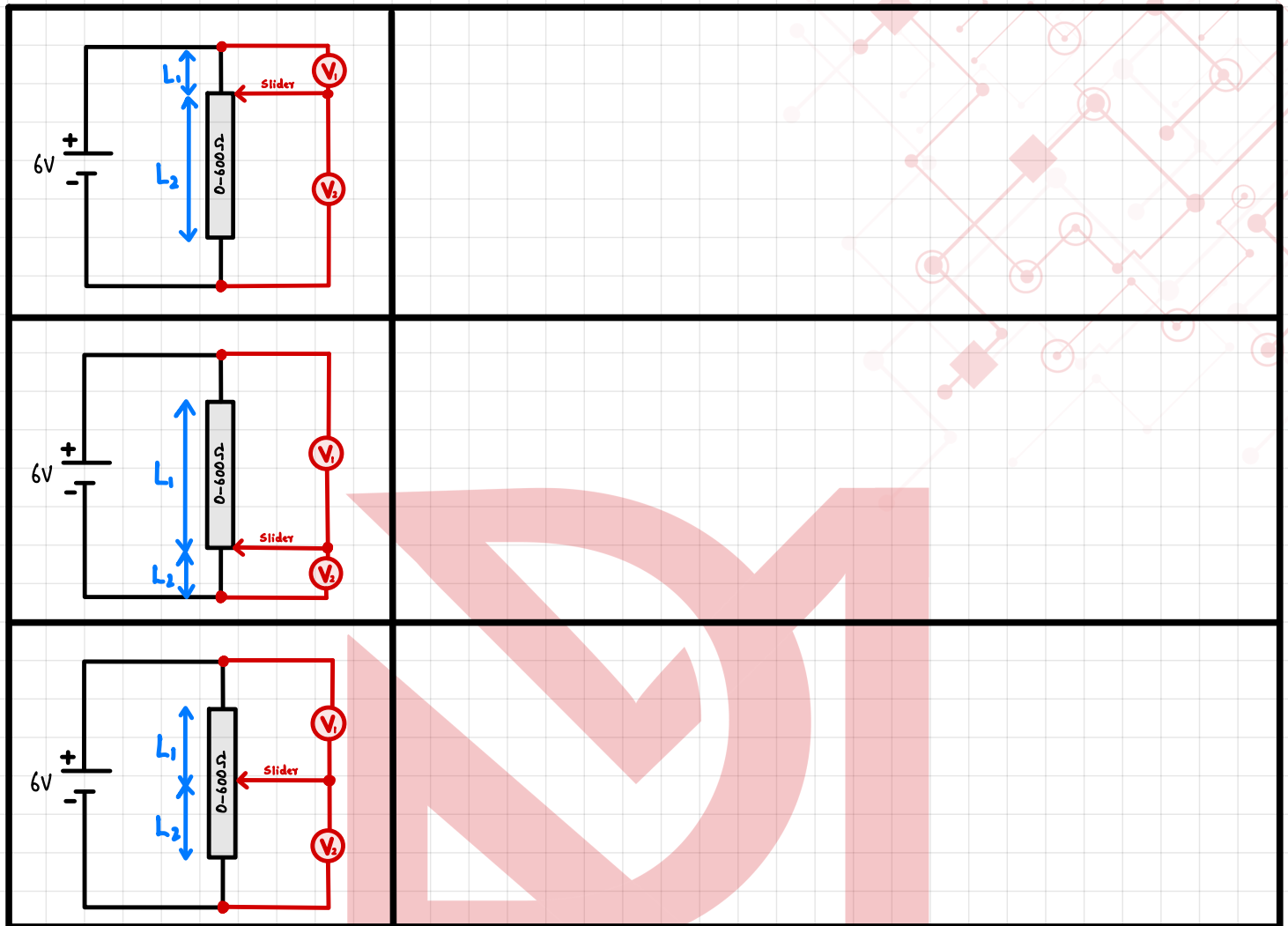
|                                                                                    |                         |                                                                                     |                         |
|------------------------------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------|-------------------------|
|   | $R_1 : R_2 = V_1 : V_2$ |   | $R_1 : R_2 = V_1 : V_2$ |
|  | $R_1 : R_2 = V_1 : V_2$ |  | $R_1 : R_2 = V_1 : V_2$ |

FIND THE VALUE OF RESISTANCE "R<sub>1</sub>" & "R<sub>2</sub>".

|                                                                                     |                         |                                                                                      |                         |
|-------------------------------------------------------------------------------------|-------------------------|--------------------------------------------------------------------------------------|-------------------------|
|  | $R_1 : R_2 = V_1 : V_2$ |  | $R_1 : R_2 = V_1 : V_2$ |
|   | $R_1 : R_2 = V_1 : V_2$ |  | $R_1 : R_2 = V_1 : V_2$ |

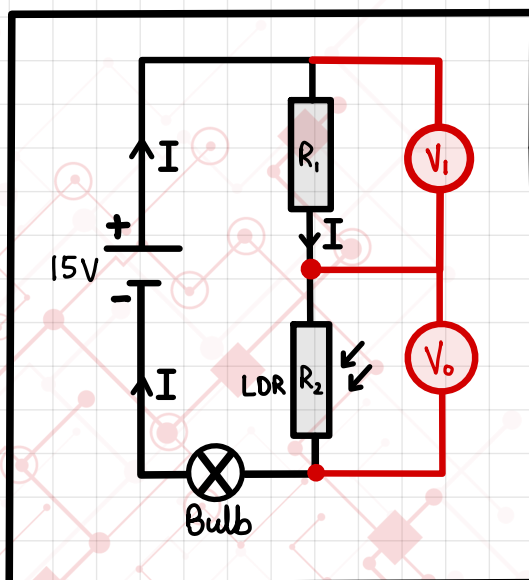


# POTENTIAL DIVIDER WITH VARIABLE RESISTOR:



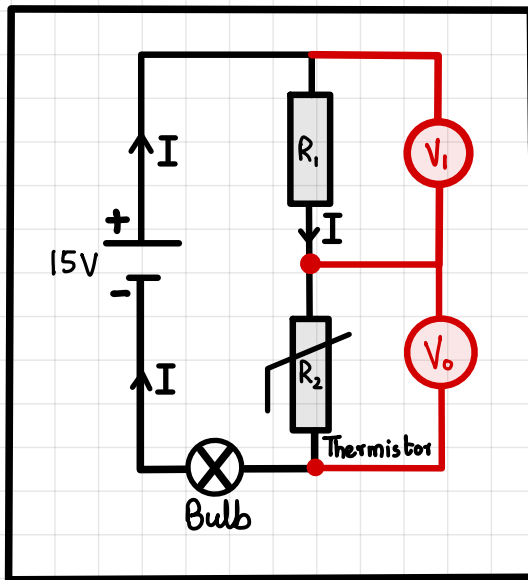
## ADDITIONAL POTENTIAL DIVIDERS:

- Fixed resistor & LDR.



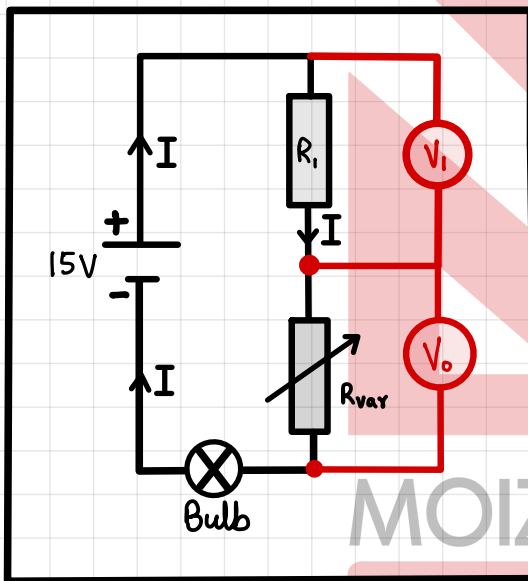
| Light | $R_{LDR}$ | $I$ | $V_0$ | $V_1$ | Bulb |
|-------|-----------|-----|-------|-------|------|
| ↑     |           |     |       |       |      |
| ↓     |           |     |       |       |      |

## • Fixed resistor & Thermistor.



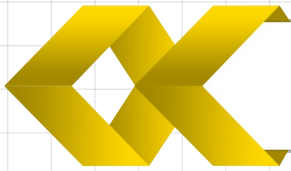
| Heat | $R_T$ | $I$ | $V_o$ | $V_i$ | Bulb |
|------|-------|-----|-------|-------|------|
| ↑    |       |     |       |       |      |
| ↓    |       |     |       |       |      |

## • Fixed resistor & Variable resistor.



| $R_1$ | $R_{var}$ | $I$ | $V_o$ | $V_i$ | Bulb |
|-------|-----------|-----|-------|-------|------|
| Same  | ↑         |     |       |       |      |
| Same  | ↓         |     |       |       |      |

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# Match the following



SCAN ME



MATCH THE FOLLOWING  
TERMINOLOGIES WITH CORRECT  
INFORMATION.

In series

only one path for electric current.

Combined resistance

in parallel.

In parallel

current remains same.

Current divides

is equal to sum of resistors in series.

Series circuits have

multiple paths to flow current.

Parallel circuits have

voltage remains same.

Thermistor works

conductor in bright light.

Potential dividers

in series circuits.

Voltage dividers

are series circuits.

LDR works as

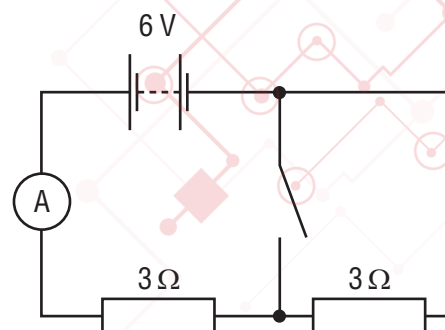
as an insulator in cold.



01 The diagram shows a circuit.

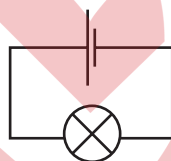
What is the reading on the ammeter when the switch is open, and the reading when it is closed?

|   | ammeter reading<br>when open / A | ammeter reading<br>when closed / A |
|---|----------------------------------|------------------------------------|
| A | 1                                | 1                                  |
| B | 1                                | 2                                  |
| C | 2                                | 1                                  |
| D | 2                                | 2                                  |

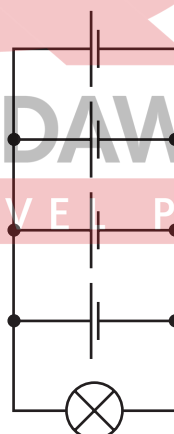


[MJ2011/P11/□ □0]

02 A lamp, designed to work at 1.5 V, is connected to a cell of electromotive force (e.m.f.) 1.5 V. The lamp lights at normal brightness.



The lamp is now connected to four similar cells, each of e.m.f. 1.5 V, arranged in parallel.

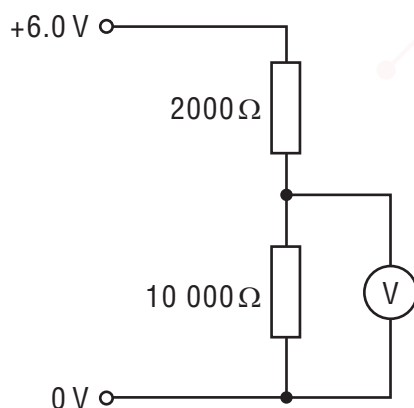


What is the effect of connecting the extra cells in this way?

- A The lamp burns out.
- B The lamp is dimmer.
- C The lamp produces light for a longer time.
- D The lamp produces light for a shorter time.

[ON2011/P11/Q29]

03 The diagram shows a potential divider system of two resistors connected to a 6.0 V power supply.

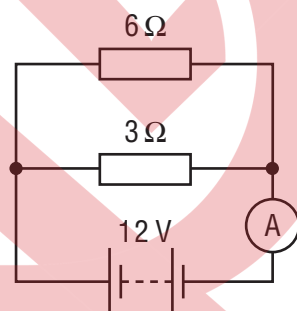


What is the reading on the voltmeter?

- A** 1.0 V      **B** 1.2 V      **C** 3.0 V      **D** 5.0 V

[ON2011/P11/Q32]

04 A 12 V battery is connected across a parallel arrangement of two resistors.



What is the reading on the ammeter?

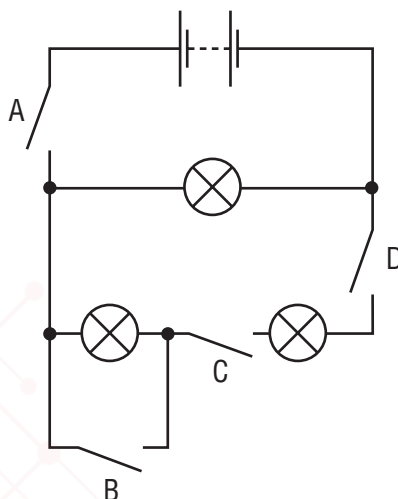
- A** 1.3 A      **B** 2.0 A      **C** 4.0 A      **D** 6.0 A

[MJ2012/P11/Q29]

05 The circuit shown is set up and then all the switches are closed.

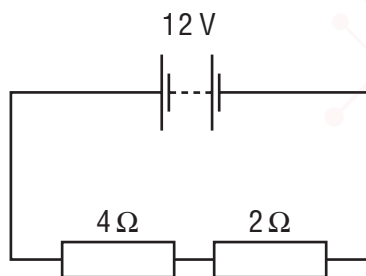
One switch is now opened and all the lamps are lit.

Which switch is opened?



[MJ2012/P11/Q30]

- 06 In the circuit shown, the potential difference (p.d.) across the  $4\Omega$  resistor is  $8\text{ V}$ .



What is the p.d. across the  $2\Omega$  resistor?

- A**  $4\text{ V}$                       **B**  $6\text{ V}$                       **C**  $8\text{ V}$                       **D**  $16\text{ V}$  [MJ2012/P12/Q30]

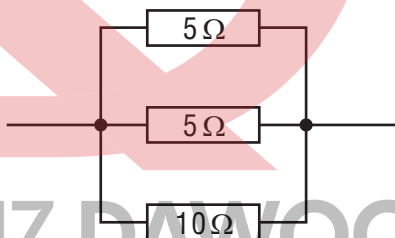
- 07 Three identical cells are connected in parallel to a resistor.

What is the advantage of using three cells in parallel, rather than a single cell?

- A** Each cell produces more energy.  
**B** Each cell supplies more charge.  
**C** Each cell takes longer to run down.  
**D** The total electromotive force (e.m.f.) is larger.

[ON2012/P11/Q28]

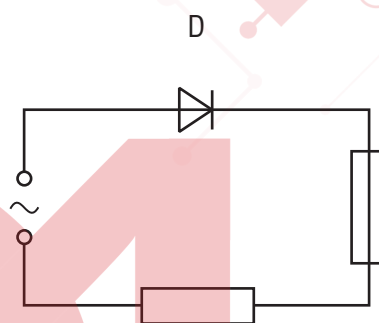
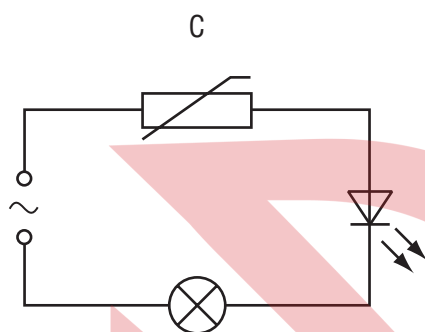
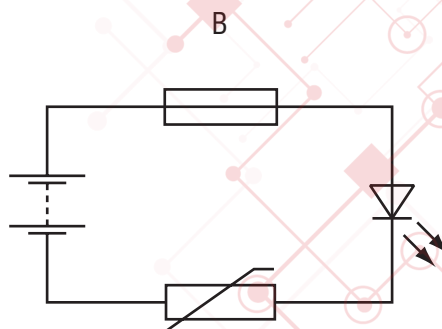
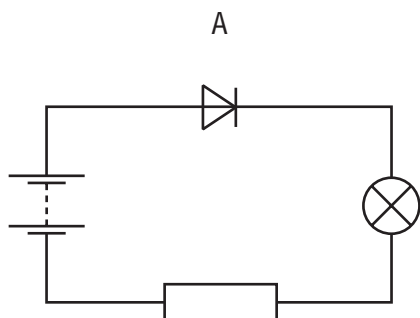
- 08 The diagram shows three resistors in parallel.



What is the combined resistance?

- A**  $0.50\Omega$                       **B**  $2.0\Omega$                       **C**  $20\Omega$                       **D**  $250\Omega$  [ON2012/P11/Q29]

09 Which circuit contains a fuse and a rectifying diode?



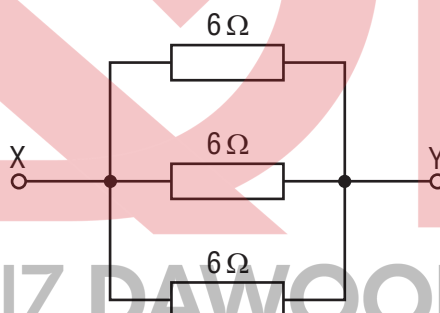
[ON2012/P11/Q30]

10 A student joins three  $6\Omega$  resistors as shown in the diagram.

Parallel ✓

Same value Resistor ✓

$$\frac{6}{3} = 2\Omega$$



What is the total resistance between points X and Y?

A  $0.5\Omega$

**B  $2\Omega$**

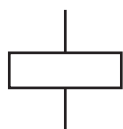
C  $6\Omega$

D  $18\Omega$

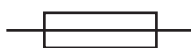
[ON2012/P12/Q29]



11 P, Q and R are electrical symbols.



P



Q



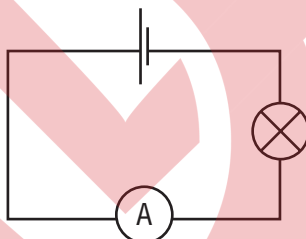
R

What do these symbols represent?

|          | P          | Q        | R          |
|----------|------------|----------|------------|
| <b>A</b> | fuse       | resistor | relay coil |
| <b>B</b> | relay coil | fuse     | resistor   |
| <b>C</b> | relay coil | resistor | fuse       |
| <b>D</b> | resistor   | fuse     | relay coil |

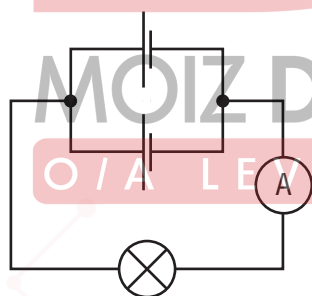
[ON2012/P12/Q30]

12 A cell is connected in series with an ammeter and a lamp. The reading on the ammeter is 1 A.

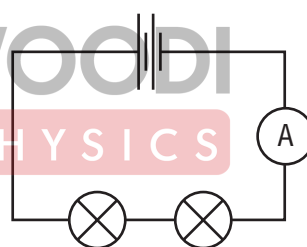


In which circuit, using identical lamps, ammeters and cells, is the reading on the ammeter 2 A?

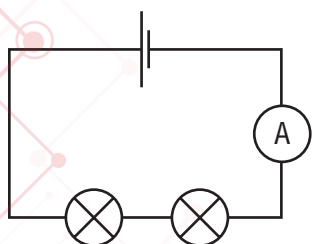
A



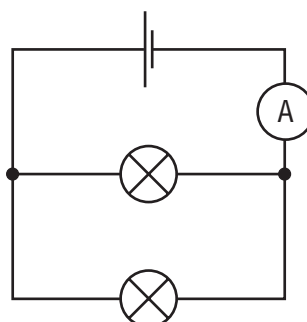
B



C

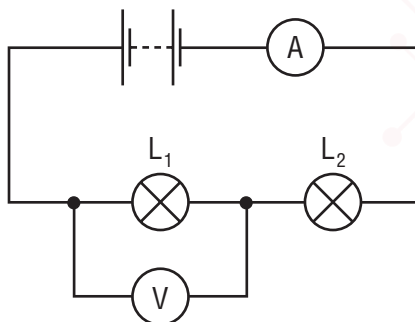


D



[MJ2013/P11/Q32]

- 13 A circuit containing two lamps  $L_1$  and  $L_2$  is connected as shown.



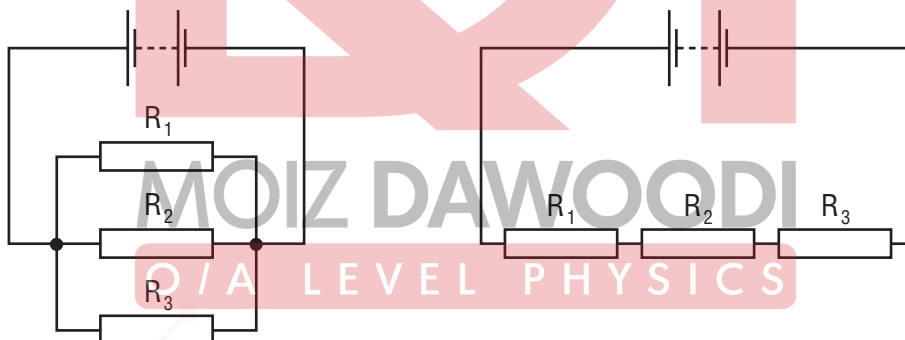
A voltmeter measures the potential difference across the lamp  $L_1$ .

The filament of lamp  $L_1$  breaks. What happens to the readings of the ammeter and of the voltmeter?

|          | reading on the ammeter | reading on the voltmeter |
|----------|------------------------|--------------------------|
| <b>A</b> | decreases              | decreases                |
| <b>B</b> | decreases              | increases                |
| <b>C</b> | increases              | decreases                |
| <b>D</b> | increases              | increases                |

[MJ2013/P12/Q34]

- 14 In the circuits P and Q below, resistors  $R_1$ ,  $R_2$  and  $R_3$  have different resistances.



circuit P

circuit Q

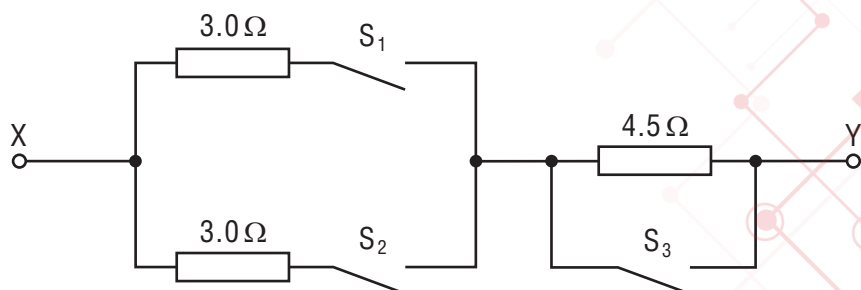
In which circuit are the currents in the resistors equal and in which circuit are the potential differences across the resistors equal?

|          | currents equal | potential differences equal |
|----------|----------------|-----------------------------|
| <b>A</b> | P              | P                           |
| <b>B</b> | P              | Q                           |
| <b>C</b> | Q              | P                           |
| <b>D</b> | Q              | Q                           |

[MJ2013/P12/Q35]



- 15 The diagram shows a circuit in which all the switches are open.

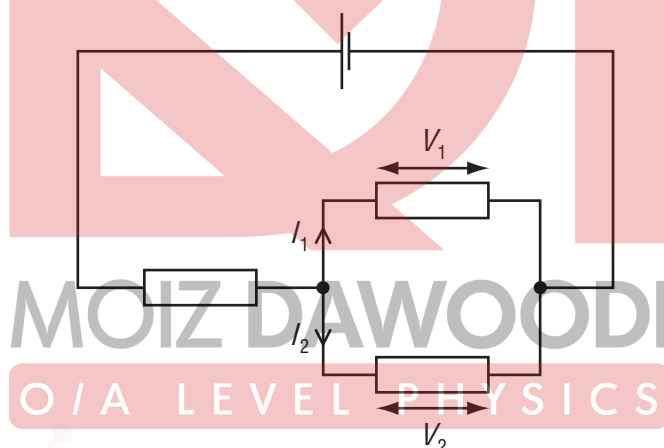


Which switch positions give a resistance of  $6.0\ \Omega$  between X and Y?

|          | $S_1$  | $S_2$  | $S_3$  |
|----------|--------|--------|--------|
| <b>A</b> | closed | closed | closed |
| <b>B</b> | closed | closed | open   |
| <b>C</b> | closed | open   | closed |
| <b>D</b> | closed | open   | open   |

[ON2013/P11/Q29]

- 16 The circuit shows three resistors connected to a cell.



The resistors have different values of resistance.

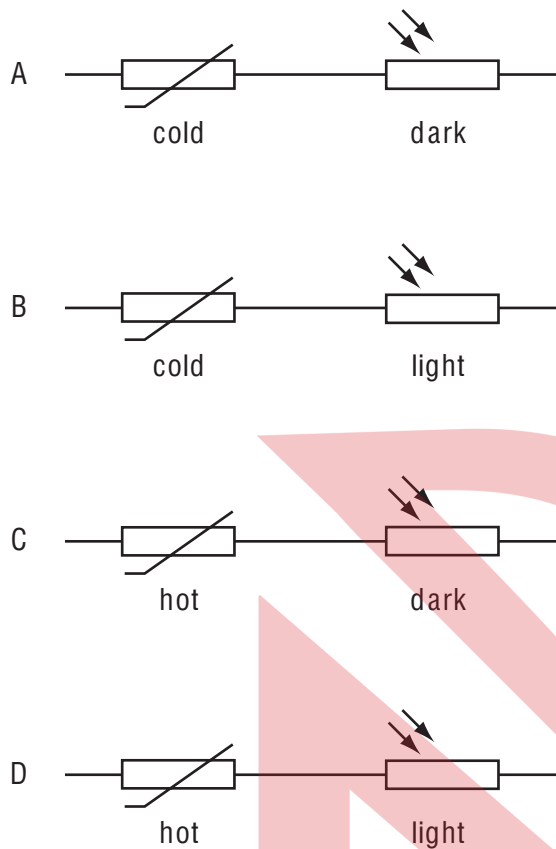
$V_1$  and  $V_2$  are potential differences and  $I_1$  and  $I_2$  are currents as shown.

Which pair of relationships is correct?

- A**  $I_1 = I_2$  and  $V_1 = V_2$
- B**  $I_1 = I_2$  and  $V_1 \neq V_2$
- C**  $I_1 \neq I_2$  and  $V_1 = V_2$
- D**  $I_1 \neq I_2$  and  $V_1 \neq V_2$

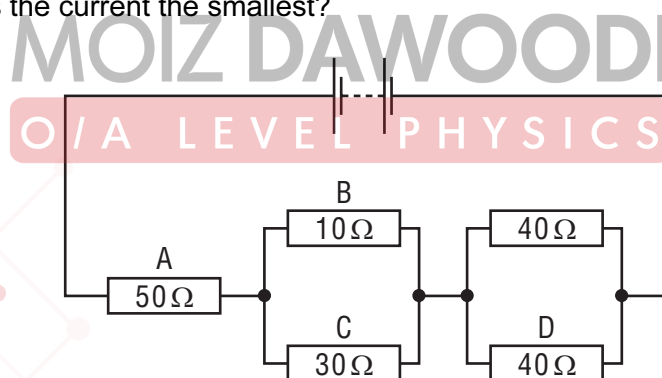
[ON2013/P11/Q30]

- 17 A thermistor and a light-dependent resistor are connected in series. Which conditions give the largest resistance?



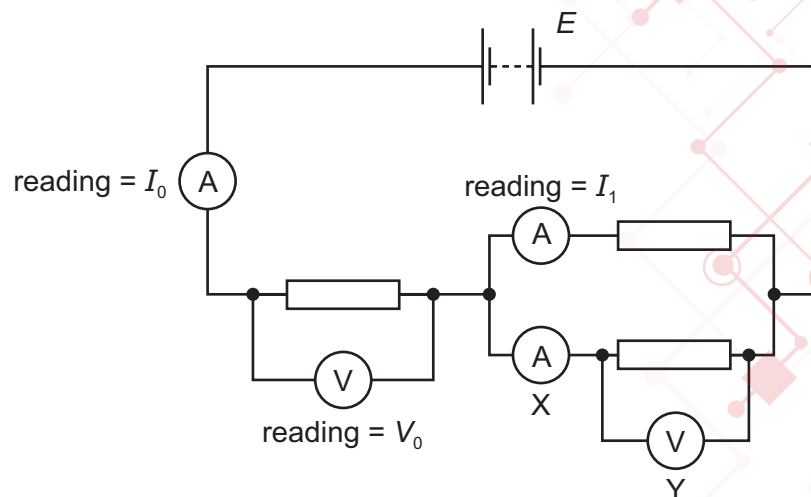
[ON2013/P11/Q36]

- 18 The diagram shows a circuit containing five resistors connected to a battery. In which resistor is the current the smallest?



[ON2013/P12/Q31]

- 19 A battery of e.m.f.  $E$  is connected in a circuit containing three resistors.



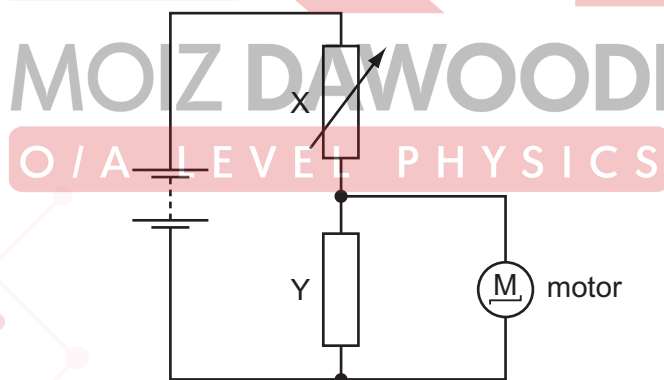
The readings of two ammeters and a voltmeter are shown.

Which readings are shown on ammeter X and on voltmeter Y?

|          | X           | Y         |
|----------|-------------|-----------|
| <b>A</b> | $I_0 - I_1$ | $E - V_0$ |
| <b>B</b> | $I_0 - I_1$ | $E + V_0$ |
| <b>C</b> | $I_0 + I_1$ | $E + V_0$ |
| <b>D</b> | $I_0 + I_1$ | $E - V_0$ |

[MJ2014/P11/Q31]

- 20 In the circuit shown, resistors X and Y act as a potential divider to control the speed of a motor.

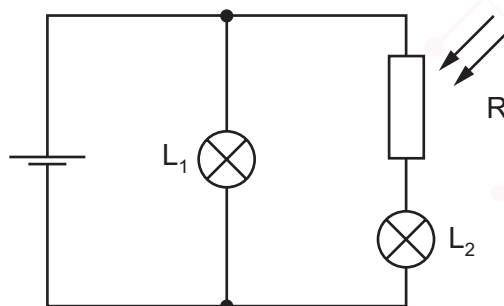


What is the reason for the potential divider?

- A** to vary the direction of the current in X
- B** to vary the e.m.f. of the battery
- C** to vary the potential difference across the motor
- D** to vary the resistance of the motor

[ON2014/P11/Q34]

- 21 In the circuit shown,  $R$  is a light-dependent resistor. Its resistance decreases when the intensity of the light shining on it increases.



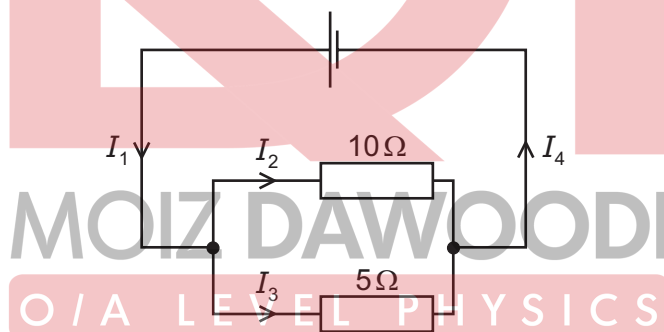
The light intensity on  $R$  increases.

What happens to the brightness of the two lamps  $L_1$  and  $L_2$ ?

|          | $L_1$          | $L_2$     |
|----------|----------------|-----------|
| <b>A</b> | decreases      | decreases |
| <b>B</b> | decreases      | increases |
| <b>C</b> | stays the same | decreases |
| <b>D</b> | stays the same | increases |

[ON2014/P11/Q35]

- 22 The currents in different parts of the circuit are  $I_1$ ,  $I_2$ ,  $I_3$  and  $I_4$ .

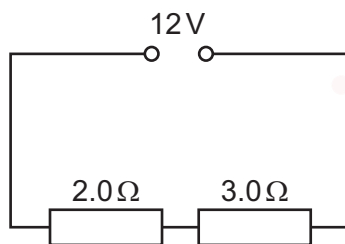


Which statement is correct?

- A**  $I_1 = I_4$  and  $I_2$  is greater than  $I_3$ .
- B**  $I_1 = I_4$  and  $I_3$  is greater than  $I_2$ .
- C**  $I_2$  is greater than  $I_1$  and less than  $I_3$ .
- D**  $I_2$  is greater than  $I_1$  and greater than  $I_3$ .

[ON2014/P12/Q25]

- 23 The diagram shows a circuit that has two resistors in series with a 12 V supply.



What is the current in the circuit?

A 2.4 A

B 10 A

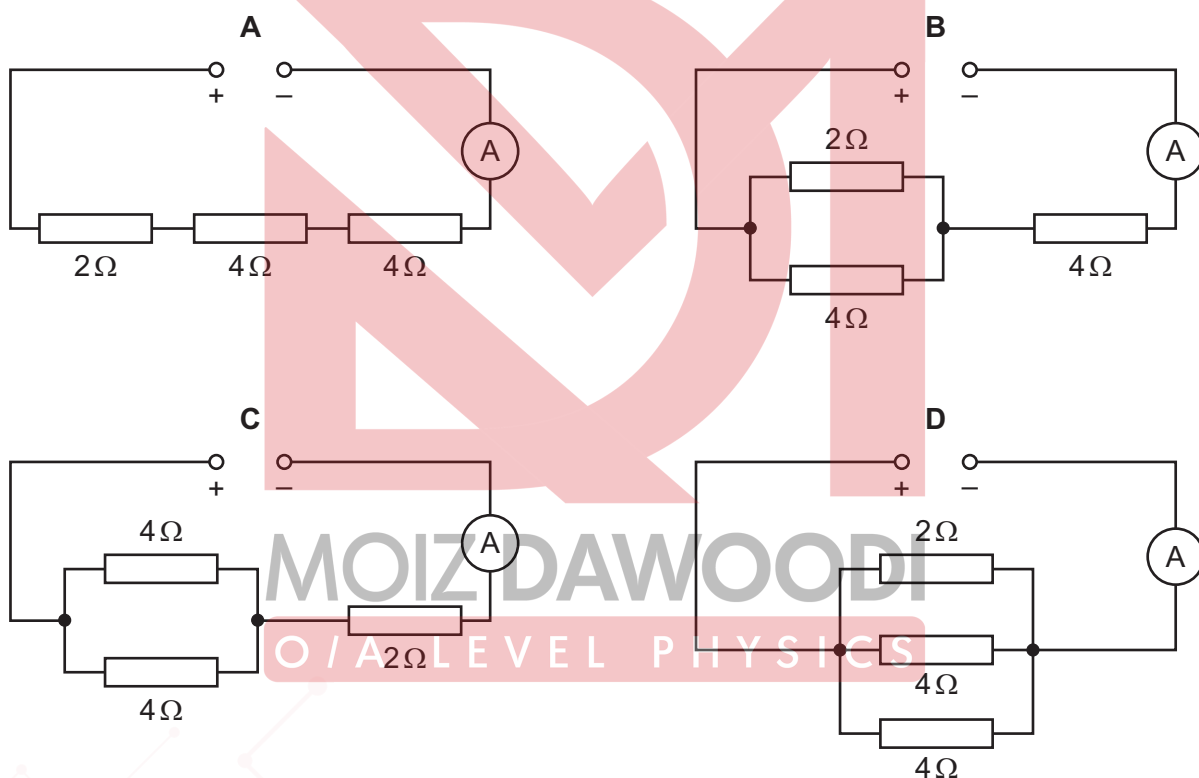
C 14 A

D 60 A

[ON2014/P12/Q26]

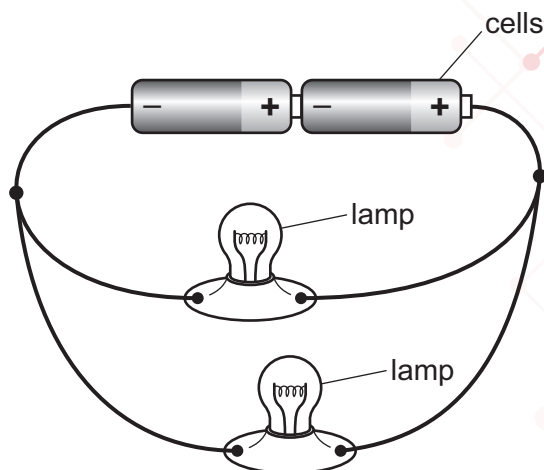
- 24 An ammeter is connected to three resistors and a power supply.

Which arrangement of resistors gives the greatest ammeter reading?

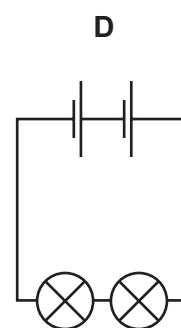
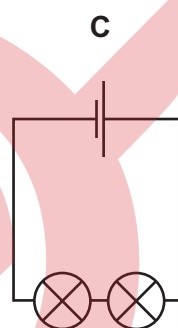
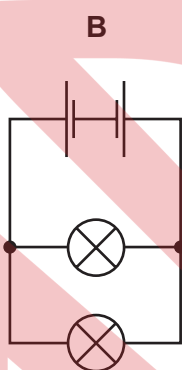
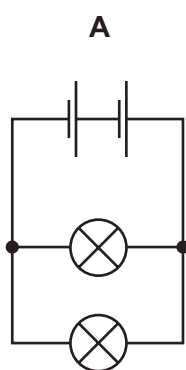


[ON2014/P12/Q27]

25 The diagram shows a circuit.



Which circuit diagram shows this circuit?



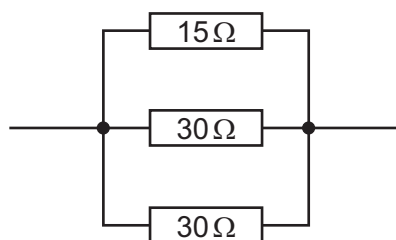
[MJ2015/P11/Q28]

26 Three identical cells are connected in parallel to a resistor. What is the advantage of using three cells in parallel, rather than using a single cell?

- A Each cell produces more energy.
- B Each cell supplies more charge.
- C Each cell takes longer to run down.
- D The total electromotive force (e.m.f.) is larger.

[MJ2015/P11/Q30]

27 The diagram shows three resistors in parallel.



What is the combined resistance?

**A**  $7.5\Omega$

**B**  $15\Omega$

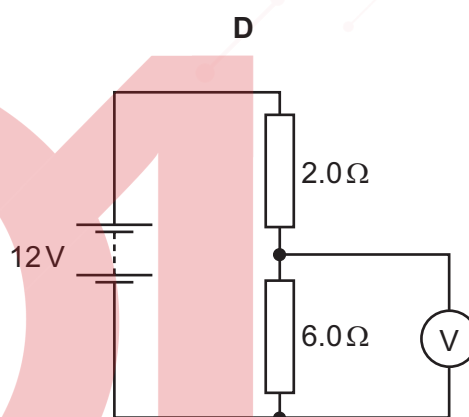
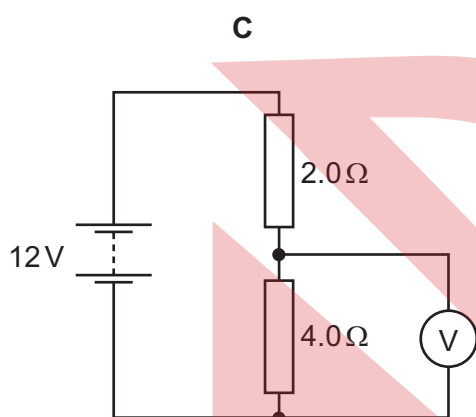
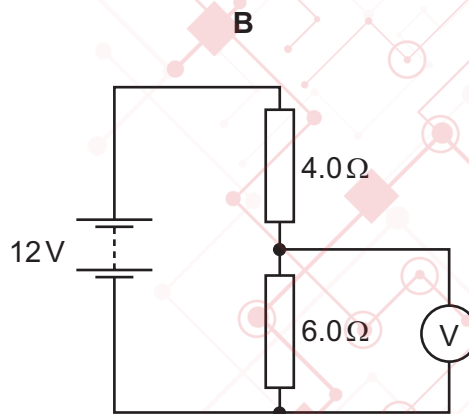
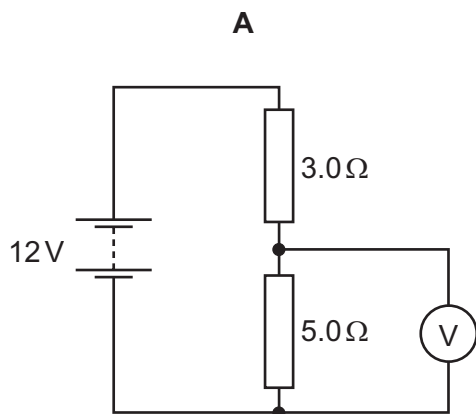
**C**  $30\Omega$

**D**  $75\Omega$

[MJ2015/P11/Q33]



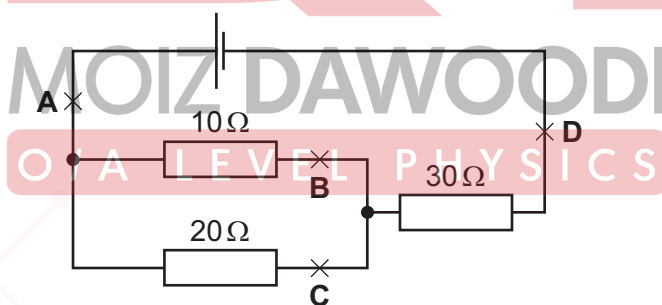
28 In which circuit is the voltmeter reading 7.2V?



[MJ2015/P11/Q34]

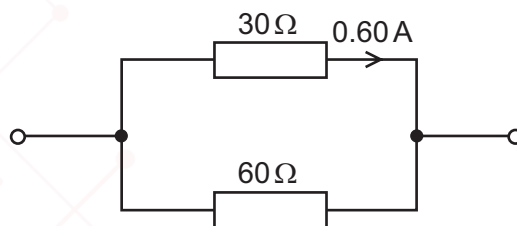
29 The diagram shows a circuit.

Where must an ammeter be connected to measure the smallest current?



[MJ2015/P12/Q33]

30 Two resistors of resistances  $30\Omega$  and  $60\Omega$  are arranged in parallel. The current in the  $30\Omega$  resistor is  $0.60\text{ A}$ .



What is the potential difference across the  $60\Omega$  resistor?

**A**  $9.0\text{ V}$

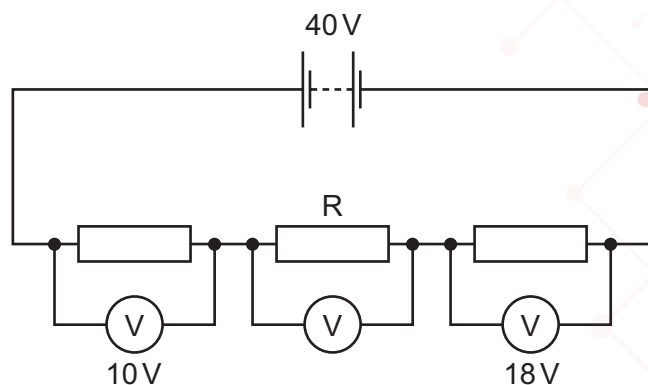
**B**  $18\text{ V}$

**C**  $36\text{ V}$

**D**  $54\text{ V}$

[MJ2015/P12/Q34]

- 31 The circuit shows three resistors in series connected to a battery. Each resistor has a voltmeter across it and two of the voltages are shown.



What is the potential difference (p.d.) across the resistor R?

- A 12V      B 22V      C 30V      D 68V

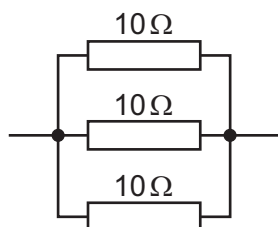
[MJ2015/P12/Q35]

- 32 Which physical quantity is produced by a calculation where a charge is multiplied by a potential difference (p.d.)?

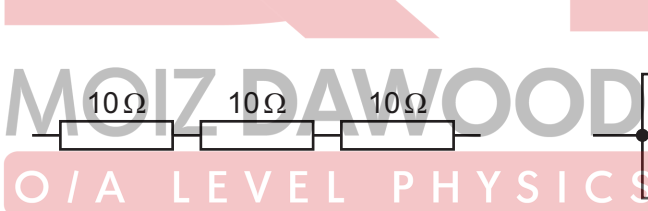
- A current  
B electromotive force (e.m.f.)  
C energy  
D power

[ON2015/P11/Q27]

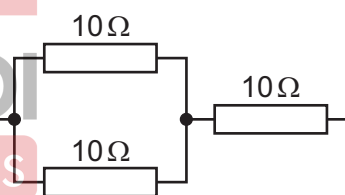
- 33 Three  $10\Omega$  resistors are connected together to form networks X, Y and Z.



network X



network Y

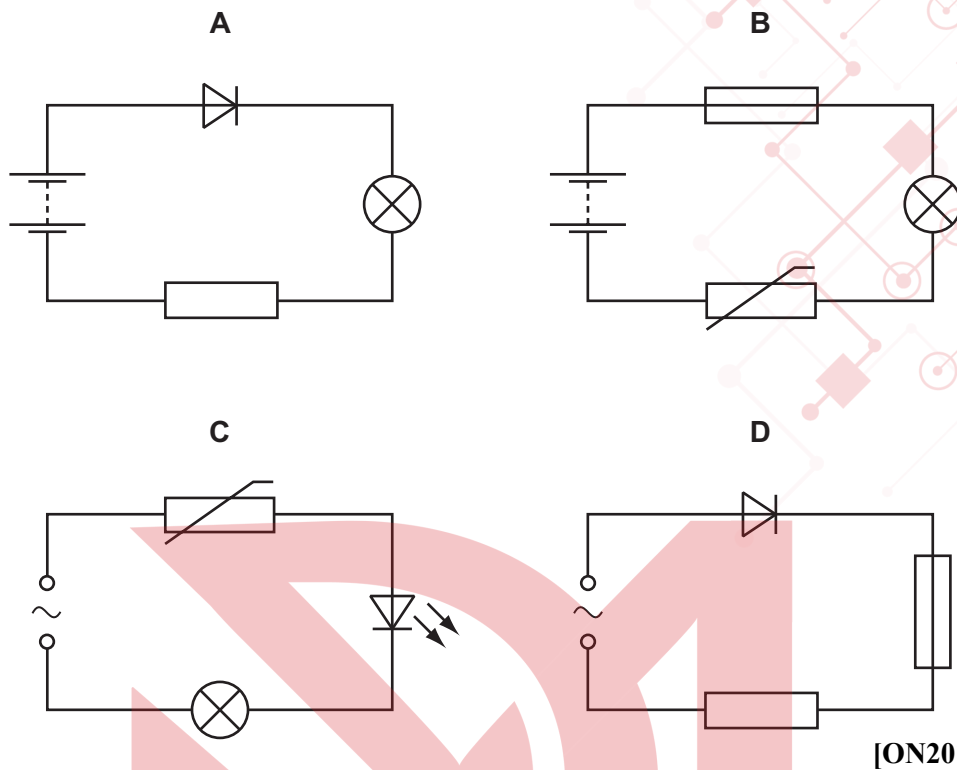


network Z

What is the order of resistance of the networks going from the smallest total resistance to the largest total resistance?

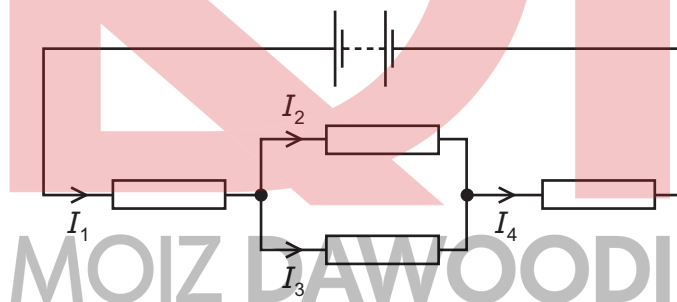
- A  $X \rightarrow Y \rightarrow Z$       B  $X \rightarrow Z \rightarrow Y$       C  $Z \rightarrow X \rightarrow Y$       D  $Z \rightarrow Y \rightarrow X$       [ON2015/P11/Q28]

34 Which circuit contains a fuse and a diode?



[ON2015/P11/Q29]

35 Four resistors of unequal resistances are connected to a power supply as shown.



The currents in the four resistors are  $I_1$ ,  $I_2$ ,  $I_3$  and  $I_4$ .

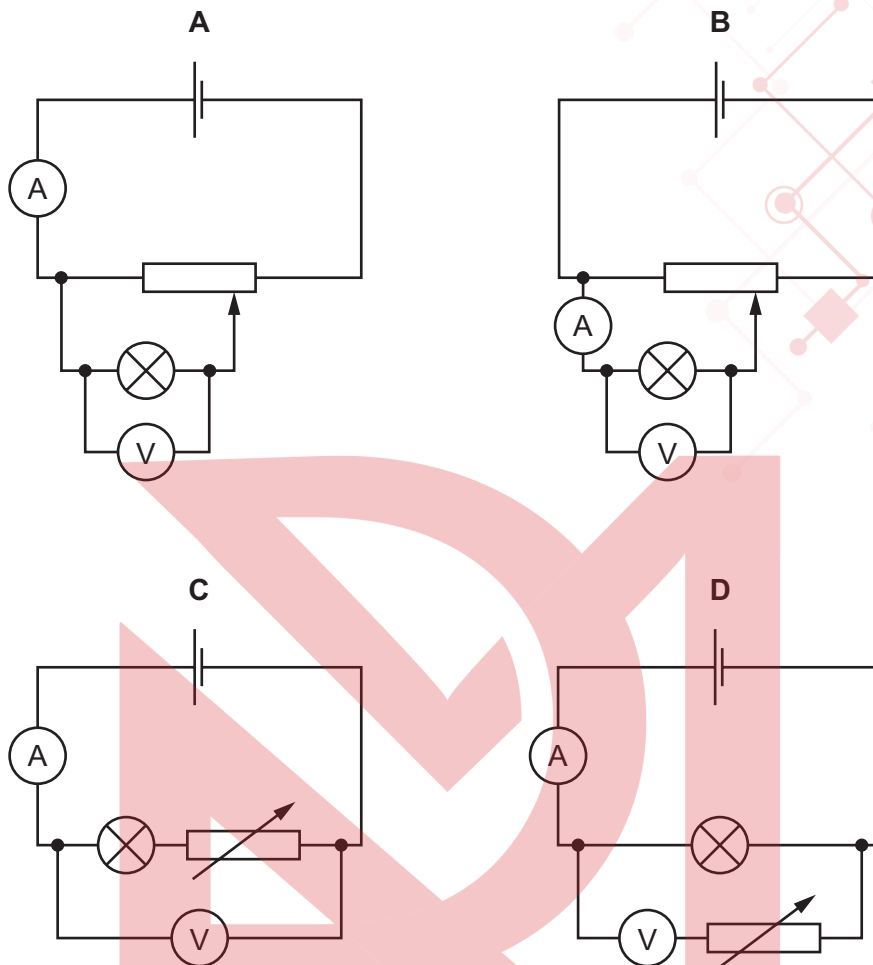
Which equation is correct?

- A  $I_1 = I_2 - I_3$
- B  $I_2 = I_1 + I_4$
- C  $I_3 = I_4 - I_1$
- D  $I_4 = I_2 + I_3$

[ON2015/P11/Q30]

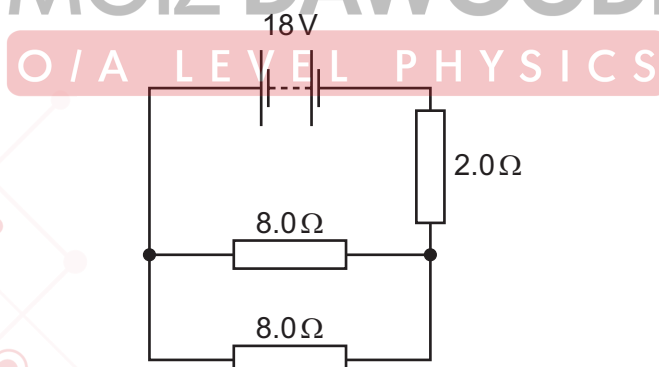
- 36 An experiment is set up to investigate how the current in a filament lamp changes with the potential difference across it.

Which circuit is correct?



[ON2015/P11/Q34]

- 37 A power supply of 18V is connected to three resistors, as shown.



What is the potential difference across the 2.0  $\Omega$  resistor?

- A 2.0V      B 3.6V      C 6.0V      D 12V

|    |   |    |   |
|----|---|----|---|
| 1  | B | 21 | D |
| 2  | C | 22 | B |
| 3  | D | 23 | A |
| 4  | D | 24 | D |
| 5  | B | 25 | A |
| 6  | A | 26 | C |
| 7  | C | 27 | A |
| 8  | B | 28 | B |
| 9  | D | 29 | C |
| 10 | B | 30 | B |
| 11 | B | 31 | A |
| 12 | D | 32 | C |
| 13 | B | 33 | B |
| 14 | C | 34 | D |
| 15 | B | 35 | D |
| 16 | C | 36 | B |
| 17 | A | 37 | C |
| 18 | C |    |   |
| 19 | A |    |   |
| 20 | C |    |   |

**MOIZ DAWOODI**  
O / A LEVEL PHYSICS

- 1 Fig. 1.1 shows part of a low-voltage lighting circuit.

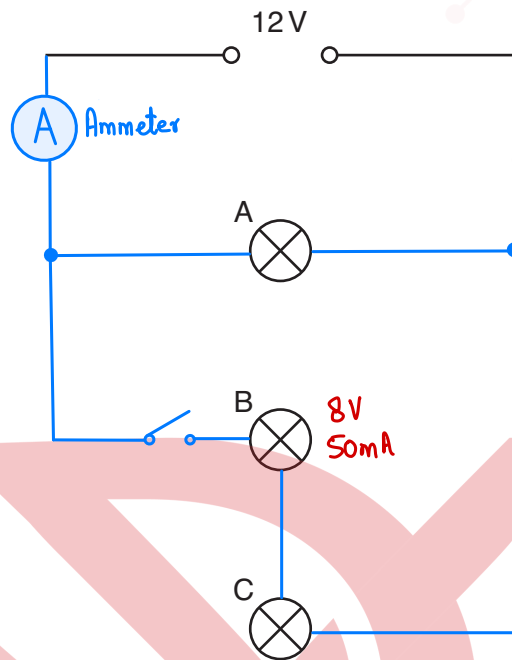


Fig. 1.1

The power supply voltage is 12V.

- (a) On Fig. 1.1, complete the circuit, adding components as necessary, so that:

- the total current in the circuit can be measured,
- lamp A is on all the time,
- lamps B and C are in series with each other and are switched on or off together.

[3]

- (b) All the lamps are on. The potential difference (p.d.) across lamp B is 8.0V and the current in lamp B is 50mA.

- (i) Calculate the resistance of lamp B.

$$R = \frac{V}{I} = \frac{8}{50 \times 10^{-3}} = 160 \, \Omega$$

resistance = 160  $\Omega$  [2]

- (ii) State the current in lamp C.

current = 50 mA [1]

[MJ2011/P22/Q5]

- 2 Fig. 2.1 is the circuit diagram of a circuit set up by a student.

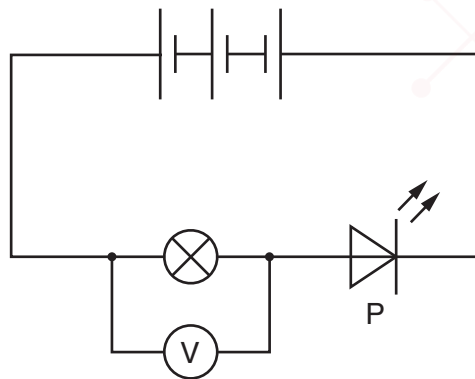


Fig. 2.1

The student uses three cells, each of electromotive force (e.m.f.) 1.5V. The cells are connected in series, as shown.

- (a) State the name of component P.

.....[1]

- (b) State what is meant by the *electromotive force* of a cell.

.....  
.....[1]

- (c) The student expects the reading on the voltmeter to be 4.5V. The actual reading is 0.2V.

Suggest **two** reasons why the voltmeter reading is much less than 4.5V.

MOIZ DAWOODI  
O/A LEVEL PHYSICS  
.....[2]

- (d) (i) In the space below, draw three cells connected in parallel.

[1]

- (ii) State one advantage of connecting cells in parallel.

.....  
.....[1]

[MJ2012/P21/Q7]



3 A student investigates how the resistance of a wire depends upon its length. The student uses an ammeter, a voltmeter, a battery, a fixed resistor and the wire under test, all connected in an electrical circuit.

(a) (i) In the space below, draw a circuit diagram of the apparatus. Label the wire under test with the letter W.

[3]

(ii) Describe how the student obtains one complete set of results.

.....

.....

.....

[2]

(iii) During the experiment, the student keeps the temperature of the wire constant.

1. Suggest why it is sensible to keep the temperature of the wire constant.

.....

.....

[1]

2. Suggest how the student keeps the temperature of the wire constant.

.....

.....

[1]

- (b) Fig. 3.1 shows part of a circuit containing three resistors X, Y and Z.

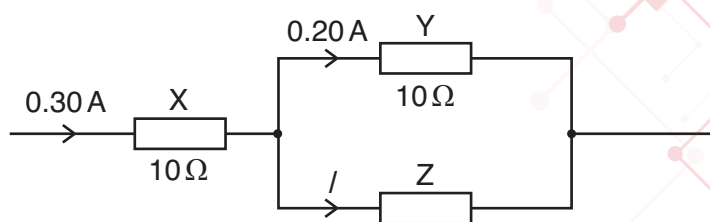


Fig. 3.1

The resistance of X and the resistance of Y are each  $10\Omega$ . The current through X is 0.30 A. The current through Y is 0.20 A.

- (i) Calculate the potential difference (p.d.) across Y.

p.d. = .....[2]

- (ii) Calculate the current  $I$  in resistor Z.

$I$  = .....[1]

- (iii) Explain in words, rather than by calculation, how Fig. 8.1 shows that the resistance of Z is larger than the resistance of Y. Use ideas about p.d. and current in your answer.

.....  
.....  
.....[2]

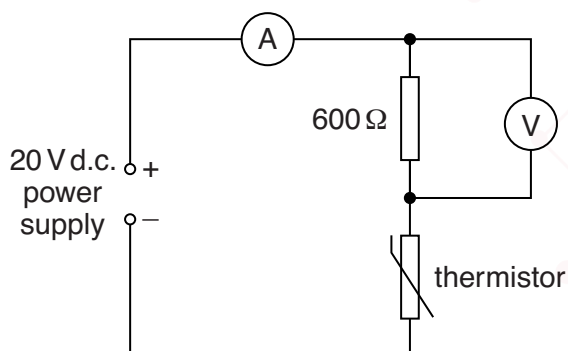
- (iv) Calculate the total resistance of the resistors X, Y and Z in this circuit.

resistance = .....[3]

[MJ2012/P22/Q8]

- 4 A  $600\Omega$  resistor and a thermistor are connected in series with an ammeter and a 20V d.c. (direct current) power supply. A voltmeter is in parallel with the resistor.

Fig. 4.1 is the circuit diagram.



**Fig. 4.1**

The ammeter reads 0.025 A.

(a) Calculate

(i) the reading on the voltmeter,

reading = ..... [2]

(ii) the resistance of the thermistor.

resistance = ..... [2]

(b) The temperature of the thermistor increases.

State what, if anything, happens to

(i) the resistance of the thermistor,

..... [1]

(ii) the ammeter reading and to the voltmeter reading.

ammeter reading: .....

voltmeter reading: .....

[1]

[ON2012/P21/Q8]

- 5 Fig. 5.1 shows the current-voltage graph for a filament lamp.

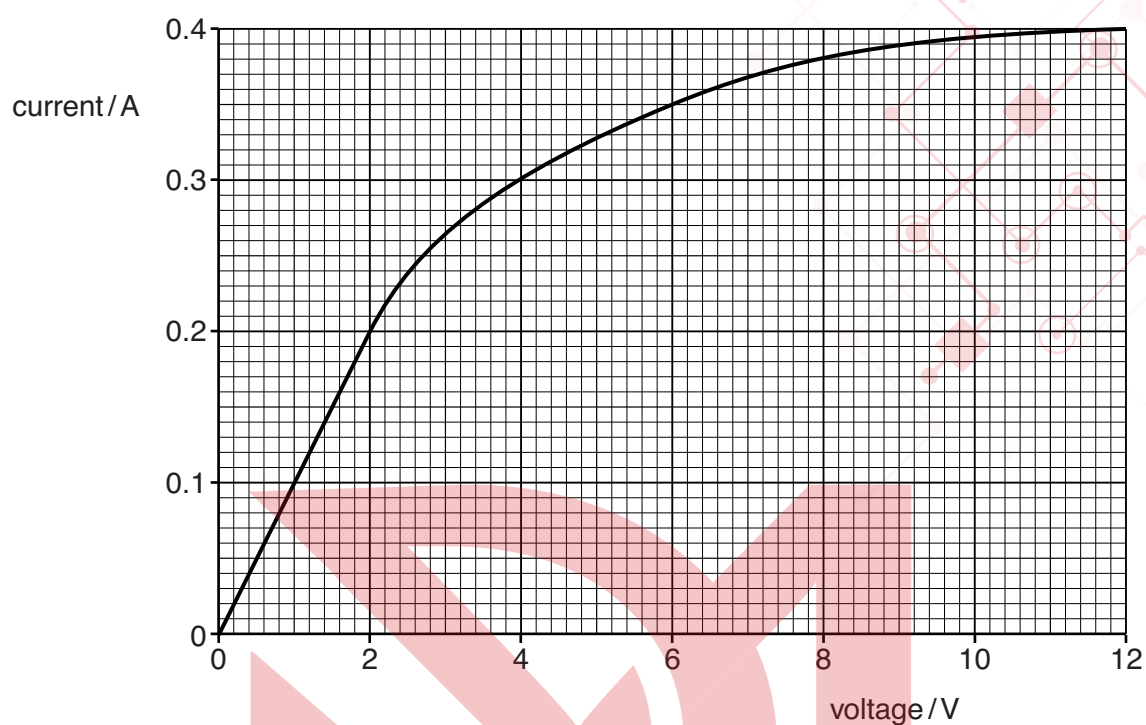


Fig. 5.1

- (a) (i) State the range of voltages where the resistance increases.

..... [1]

- (ii) State why the resistance of the lamp increases as the voltage increases.

.....

..... [1]

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- (b) The filament lamp is connected in the circuit shown in Fig. 5.2.

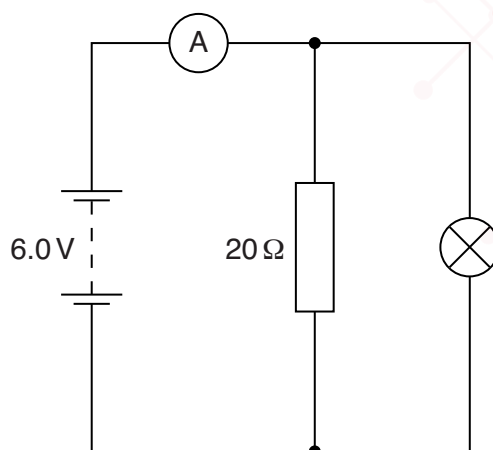


Fig. 5.2

There are currents in the lamp, the  $20\ \Omega$  resistor and the ammeter.

- (i) State what is meant by an *electric current*.

.....  
..... [1]

- (ii) Use Fig. 5.1 to determine the current in the filament lamp.

current = ..... [1]

- (iii) Determine the current in the ammeter.

current = ..... [3]

[MJ2013/P22/Q6]

- 6 (a) Fig. 6.1 shows a variable potential divider (potentiometer) connected to a 6.0 V battery.

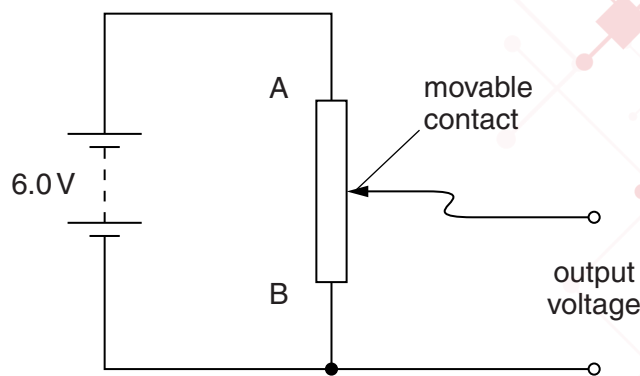


Fig. 6.1

The movable contact can slide between A and B. As it moves, the output voltage changes.

Complete the table below to show the output voltage when the contact is at A and at B.

| position of movable contact | A | B |
|-----------------------------|---|---|
| output voltage              |   |   |

[2]

- (b) Fig. 6.2 shows a thermistor in a potential divider circuit that is used to monitor temperature. The fixed resistor has a resistance of  $2000\ \Omega$

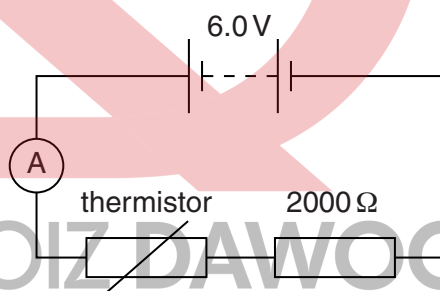


Fig. 6.2

- (i) Explain why the voltage across the thermistor decreases as its temperature rises.

.....  
 ..... [2]

- (ii) At one temperature, the thermistor has a resistance of  $1000\ \Omega$ . Calculate the current in the thermistor.

current = ..... [2]

[MJ2014/P21/Q7]

- 7 A student sets up the circuit shown in Fig. 7.1.

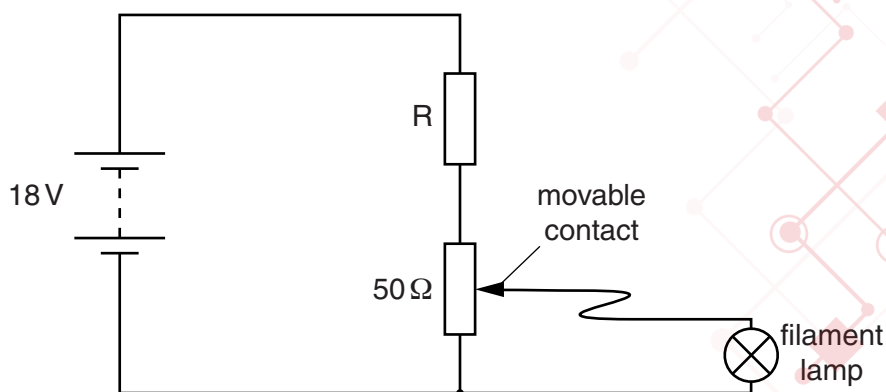


Fig. 7.1

R is a fixed resistor in the circuit. The filament lamp is marked 12V, 0.25A.

The circuit is used to produce a current/voltage graph for the filament lamp. The ammeter and voltmeter needed are not shown.

To obtain different readings, the student changes the position of the movable contact.

- (a) On Fig. 7.1, draw the symbols for an ammeter and a voltmeter in the correct positions. [3]

- (b) Explain why it is sensible to include the resistor R in this circuit.

.....

.....

..... [2]

- (c) (i) On Fig. 7.2, sketch a current/voltage graph for the lamp.

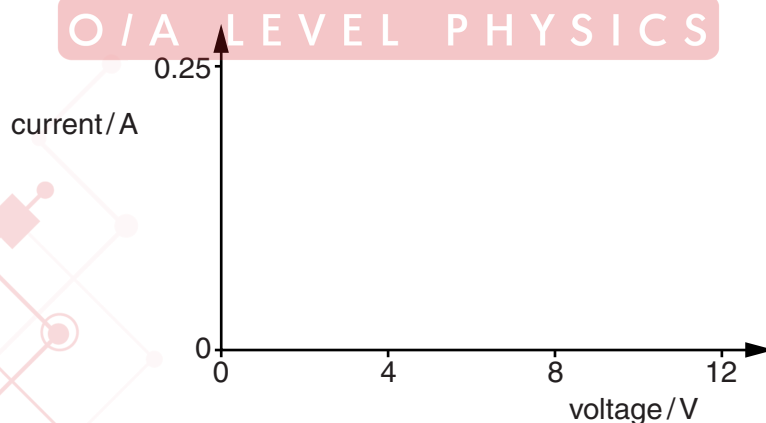


Fig. 7.2

[3]



- (ii) State and explain how a current/voltage graph for a fixed resistor is different from the graph for a filament lamp.

.....  
 .....  
 ..... [2]

- (d) Fig. 7.3 shows the position of the movable contact when the voltage across the lamp is 12 V and the current in the lamp is 0.25 A.

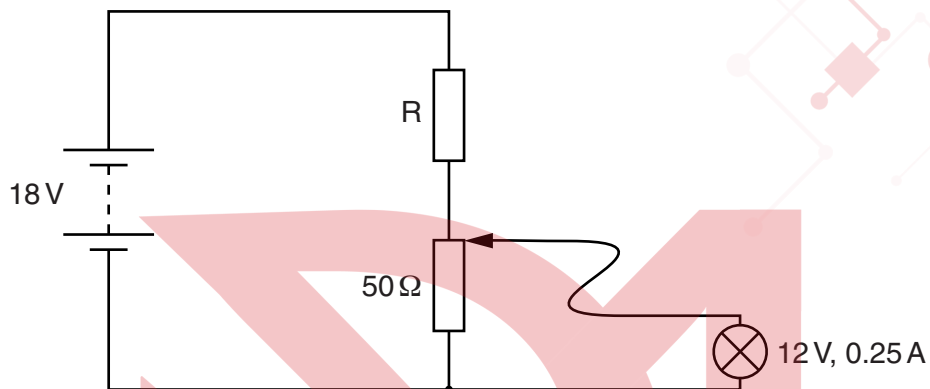


Fig. 7.3

Determine

- (i) the current in the  $50\Omega$  resistor,

current = ..... [2]

- (ii) the current in R,

current = ..... [1]

- (iii) the potential difference (p.d.) across R,

p.d. = ..... [1]

- (iv) the resistance of R.

resistance = ..... [1]

[MJ2014/P22/Q11]

- 8 A student makes a 2.0V battery by connecting two cells of electromotive force (e.m.f.) 2.0V in parallel. The battery, an ammeter with different ranges and three different resistors are used to set up the circuit shown in Fig. 8.1.

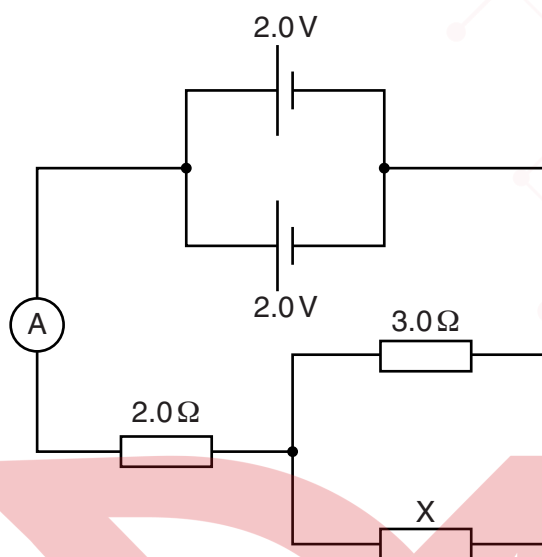


Fig. 8.1

- (a) (i) Explain what is meant by *electromotive force*.

.....  
 .....  
 .....[2]

- (ii) State one advantage of using two cells in parallel rather than using a single 2.0V cell.

.....  
 .....[1]

- (b) Resistor X and the 3.0Ω resistor have a combined resistance that is equal to 2.0Ω.

Calculate

- (i) the total resistance of the circuit,

total resistance = .....[1]

- (ii) the resistance of X.

resistance of X = .....[2]

- (c) (i) Determine the reading of the ammeter.

reading = .....[2]

- (ii) Suggest a suitable range for the ammeter.

.....[1]

- (d) The current in the  $2.0\Omega$  resistor is  $I_2$ .  
The current in the  $3.0\Omega$  resistor is  $I_3$ .  
The current in X is  $I_X$ .

State the equation that relates  $I_2$ ,  $I_3$  and  $I_X$ .

.....[1]

- (e) State the potential difference (p.d.) across

- (i) the  $2.0\Omega$  resistor,

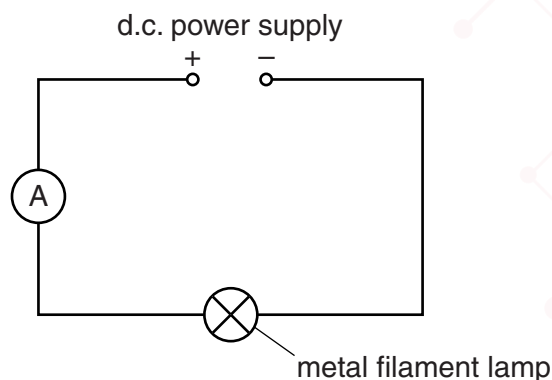
p.d. = .....[1]

- (ii) the  $3.0\Omega$  resistor.

p.d. = .....[1]

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- (f) The student sets up a second circuit using a variable d.c. power supply, an ammeter and a 12V metal filament lamp. The circuit is shown in Fig. 8.2.



**Fig. 8.2**

The d.c. power supply is set to 12V and the ammeter reading is 1.5A.

The student changes the e.m.f. of the d.c. power supply to 6.0V. The lamp dims and the ammeter reading changes.

- (i) State and explain what happens to the resistance of the metal filament of the lamp.

.....  
.....  
.....[2]

- (ii) State whether the new ammeter reading is less than, equal to or greater than 0.75 A.

.....[1]

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[ON2014/P21/Q11]

- 9 Fig. 9.1 shows a circuit that contains a resistor connected to a power supply of 6.0V and a lamp L.

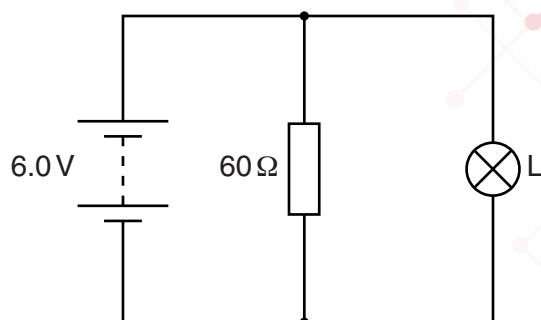


Fig. 9.1

The resistor has a resistance of  $60\Omega$ . The lamp is marked 6.0V, 0.90W.

(a) Calculate

(i) the current in the resistor,

current = .....[2]

(ii) the current in the power supply.

current = .....[2]

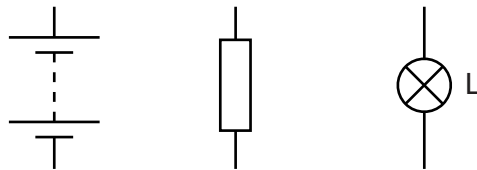
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(b) A second lamp is added to the circuit shown in Fig. 9.1.

The second lamp is in series with the  $60\Omega$  resistor, but is **not** in series with lamp L.

(i) In the space below draw a circuit diagram of this new circuit.

The power supply,  $60\Omega$  resistor and lamp L have been drawn for you.



[1]

(ii) The two lamps are identical.

Explain why the second lamp is dimmer than lamp L.

A calculation is not required.

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[1]

[MJ2015/P21/Q6]

- 10 A potential divider is made from a light-dependent resistor (LDR) and a  $6.0\text{ k}\Omega$  fixed resistor. The potential divider is connected in series with a  $12\text{ V}$  d.c. power supply and a voltmeter is connected across the  $6.0\text{ k}\Omega$  resistor. Fig. 10.1 is the circuit diagram.

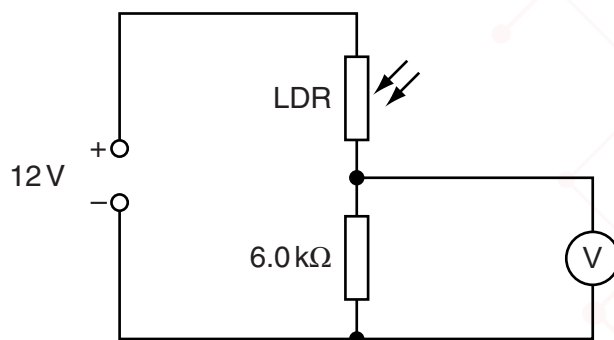


Fig. 10.1

A light shines on the LDR. The resistance of the LDR is  $2.0\text{ k}\Omega$ .

(a) Calculate

(i) the current in the LDR,

current = ..... [2]

(ii) the reading on the voltmeter.

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reading = ..... [1]

(b) The brightness of the light on the LDR is gradually increased.

State and explain what happens to the reading on the voltmeter.

.....  
.....  
..... [2]

(c) Suggest a use for a potential divider made from an LDR and a fixed resistor.

.....  
..... [1]

[ON2015/P21/Q8]



| Page 3 | Mark Scheme: Teachers' version | Syllabus | Paper |
|--------|--------------------------------|----------|-------|
|        | GCE O LEVEL – May/June 2011    | 5054     | 22    |

- 5 (a) ammeter in series with supply // ammeters in series with A **and** in series with B & C B1
- A across cell with no switch (condone closed switch) **not**  B1
- B and C in series with switch (closed or open) and cell B1
- (b) (i) ( $R =$ )  $V/I$  in any form numerical or algebraic, e.g.  $8/50$ ,  $8/0.05$  C1
- 160  $\Omega$  A1
- (ii) 50 mA // 0.05(0) A B1 [6]

| Page 4 | Mark Scheme: Teachers' version | Syllabus | Paper |
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|        | GCE O LEVEL – May/June 2012    | 5054     | 21    |

- 7 (a) LED; light-emitting diode B1
- (b) energy/work done per unit charge/coulomb B1
- (c) voltage/p.d. across P B1
- one cell connected the wrong way/acting against the others B1
- (d) (i) correct arrangement B1
- (ii) cells last longer; cells run down slower; one cells fails the others still work; reduces (internal) resistance; if cell removed circuit not broken B1 [6]

| Page 5 | Mark Scheme: Teachers' version | Syllabus | Paper |
|--------|--------------------------------|----------|-------|
|        | GCE O LEVEL – May/June 2012    | 5054     | 22    |

- 8 (a) (i) **correct circuit symbols** containing, in any circuit, a
- battery/cell/d.c. power supply
  - ammeter
  - voltmeter
  - fixed resistor
- B1
- ammeter clearly measures current through W B1
- voltmeter clearly across W if W shown or a resistor if not B1
- (ii) Any 2 from
- resistance (calculated from)  $V/I$  or  $V = IR$  seen
  - length (of wire),  $V$  and  $I$  **all three** measured
  - change length **and**  $V$  and  $I$  measured
- B2
- (iii) 1. resistance/resistivity changes (with temperature)
- OR wire gets hot **and** melts/burns/catches fire/dangerous
- OR  $V$  only proportional to  $I$  at constant temperature B1
2. use of a water bath/heat sink
- OR use small currents
- OR take reading (quickly) and switch off B1

- (b) (i) ( $V =$ )  $IR$  in any form numerical or algebraic  
2(.0)V C1  
A1
- (ii) 0.1(0)A B1
- (iii) (Z) has the same potential difference/voltage B1  
(Z) has less/small(er) current (thus larger resistance) B1
- (iv) (p.d. across X =)  $0.3 \times 10$  (V) C1  
OR ( $R_Z =$ )  $2/0.1$  OR 20 ( $\Omega$ ) seen A1 [15]
- (total p.d.) 5 (V)  
OR 6.7 ( $\Omega$ ) seen  
OR  $1/R_T = 1/R_1 + 1/R_2$  in any form numerical or algebraic OR 20/3 seen  
16.7  $\Omega$  ; 17  $\Omega$  ; 16.67  $\Omega$  ; 16.66  $\Omega$

| Page 4 | Mark Scheme                         | Syllabus | Paper |
|--------|-------------------------------------|----------|-------|
|        | GCE O LEVEL – October/November 2012 | 5054     | 21    |

- 8 (a) (i) ( $V =$ )  $IR$  or  $0.025 \times 600$  C1  
15V A1
- (ii) 5(V) or  $5/0.025$  or 800 or 800–600 C1  
200  $\Omega$  A1
- (b) (i) decreases B1
- (ii) ammeter: opposite to  
voltmeter: same as ammeter (both changes correct) B1 [6]

| Page 3 | Mark Scheme                 | Syllabus | Paper |
|--------|-----------------------------|----------|-------|
|        | GCE O LEVEL – May/June 2013 | 5054     | 22    |

- 6 (a) (i) 2 to 2.1 (V) to any value between 11 and 12 (V) B1  
or above 2/2.1(V)
- (ii) temperature increases / gets hotter B1
- (b) (i) (rate of) flow of charge/electrons B1
- (ii) 0.35 A cao B1
- (iii) ( $I =$ )  $V/R$  algebraic C1  
or 6/20  
or 6/0.35  
0.3(0) (A) C1  
or  $1/R_T = 1/20 + 1/17.1$   
or ( $R_T =$ ) 9.2 ( $\Omega$ ) seen  
0.65 A A1 [7]

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|--------|-----------------------------|----------|-------|
|        | GCE O LEVEL – May/June 2014 | 5054     | 21    |

- 7 (a) 

|        |      |
|--------|------|
| A      | B    |
| 6.(0)V | 0(V) |

 [B1, B1]
- (b) (i) resistance (of thermistor) decreases [B1]  
current increases **and** larger voltage across 2000  $\Omega$ /fixed resistor  
**or** smaller **fraction** of voltage across thermistor **or** potential divider explained [B1]
- (ii) ( $I =$ )  $V/R$  in symbols or numbers [A1]  
0.002(0) A; 2 mA [C1]

| Page 5 | Mark Scheme                 | Syllabus | Paper |
|--------|-----------------------------|----------|-------|
|        | GCE O LEVEL – May/June 2014 | 5054     | 22    |

- 11 (a) ammeter and voltmeter correct symbols [B1]  
ammeter in series with lamp [B1]  
voltmeter in parallel with lamp [B1]
- (b) R limits or reduces the current/voltage otherwise lamp blows  
**or**  
more of the 50  $\Omega$  can be used to adjust voltage/current [B1]
- (c) (i) 12 V, 0.25 A correctly plotted (by eye) [B1]  
curved line from origin [B1]  
correct curvature – decreasing slope [B1]
- (ii) straight line (for fixed resistor) [B1]  
**lamp** has changing temperature or changing resistance  
**or** fixed resistor has constant temperature or constant **resistance** [B1]
- (d) (i) ( $I =$ )  $V/R$  in any algebraic or numerical form, e.g. 12/50 [C1]  
0.24 A [A1]
- (ii) 0.49 A [B1]
- (iii) 6(.0)V [B1]
- (iv) 12(.24) $\Omega$  [B1]
- [15]**

| Page 5 | Mark Scheme                               | Syllabus | Paper |
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- 11 (a) (i) energy to drive charge around a circuit **or** terminal p.d. on open circuit B1  
energy to drive **unit** charge around a circuit **or** energy/charge B1
- (ii) lasts longer **or** lower internal resistance **or** can replace a cell without switching off **or** continues to work if one cell is flat **ignore** more current (not greater e.m.f./voltage) B1 **[3]**

- (b) (i)  $4.0\Omega$  B1
- (ii)  $(1/R_{\text{tot}} = )1/R_1 + 1/R_2$  or  $1/3 + 1/X$  or product/sum or  $(3 \times X)/(3 + X)$  or  
 $\frac{1}{X} = \frac{1}{2} - \frac{1}{3}$  C1  
 $6.0\Omega$  A1 [3]
- (i)  $(I = ) V/R$  or  $2.0/4.0$  C1  
 $0.50\text{ A}$  A1
- (c) (ii) (from) 0 and (to) 0.50 to 5.0 A B1 [3]
- (d)  $I_2 = I_3 + I_X$  B1 [1]
- (e) (i)  $1.0\text{ V}$  B1
- (ii)  $1.0\text{ V}$  B1 [2]
- (f) (i) temperature decreases B1  
resistance decreases B1
- (ii) greater than  $0.75\text{ A}$  (e.c.f. resistance increases in (f)(i)) B1 [3]

[Total: 15]

| Page 3 | Mark Scheme                       | Syllabus | Paper |
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- 6 (a) (i)  $(I = ) V/R$  or  $6/60$  C1  
 $0.1(0)\text{ A}$  A1
- (ii)  $(I = ) P/V$  or  $0.9/6$  C1  
or  $0.15\text{ (A) seen}$   
 $0.25\text{ A}$  A1
- (b) (i) lamp correctly drawn in series with resistor but not the lamp B1
- (ii) less voltage (across lamp) because some voltage across resistor/shares voltage with resistor B1  
or less current because of effect of resistor

| Page 3 | Mark Scheme                               | Syllabus | Paper |
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- 8 (a) (i)  $(I = ) V/R$  or  $12/(6000 + 2000)$  or  $12/8000$  or  $12/2000$  or  $12/6000$  C1  
or in (ii)  $(V = ) IR$  or  $0.0015 \times 6000$  A1  
 $1.5\text{ mA}$
- (ii)  $9.0\text{ V}$  B1
- (b) (reading/it) increases B1  
resistance of LDR falls B1
- (c) light meter/sensor or automatic light switch or something sensible B1 [6]